Information and Communications University

School of Humanities

(Department of Economics)

STUDY MANUAL

2012

INTRODUCTION TO MACROECONOMICS

CHAPTER 1

NATIONAL INCOME

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After studying this chapter, the students should be able to:

- Describe the national income of a country
- Explain the relationship between output, income and expenditure using the circular flow of income
- Explain the measurement of the national income
- Discuss the problems associated with the measurement of national income
- Appreciate the uses of national income figures
- Explain the factors that determine a country's national income

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1.0 Introduction

Measures of national income and output are used in Economics to estimate the value of goods and services produced in an economy. They use a system of **national accounts** or **national accounting** first developed during the 1940s. Some of the more common measures are **Gross National Product (GNP)**, Gross Domestic Product **(GDP)**, Gross National Income **(GNI)**, Net National Product **(NNP)**, and Net National Income **(NNI)**.

There are at least three different ways of calculating these numbers. The **expenditure approach** determines aggregate demand, or Gross National Expenditure, by summing consumption, investment, government expenditure and net exports. On the other hand, the **income approach** and the closely related **output approach** can be seen as the summation of consumption, savings and taxation. The three methods must yield the same results because the total expenditures on goods and services (GNE) must by definition be equal to the value of the goods and services produced (GNP) which must be equal to the total income paid to the factors that produced these goods and services (GNI).

In practice, there are minor differences in the results obtained from the various methods due to changes in inventory levels. This is because goods in inventory have been produced (and therefore included in GDP), but not yet sold (and therefore not yet included in GNE). Similar timing issues can also cause a slight discrepancy between the value of goods produced (GDP) and the payments to the factors that produced the goods, particularly if inputs are purchased on credit.

Gross National Product

Gross National Product (GNP) is the total value of final goods and services produced in a year by a country's nationals (including profits from capital held abroad).

Gross Domestic Product

Gross Domestic Product (GDP) is the total value of final goods and services produced within a country's borders in a year.

To convert from GNP to GDP you must subtract factor income receipts from foreigners that correspond to goods and services produced abroad using factor inputs supplied by domestic sources. To convert from GDP to GNP you must add factor input payments to foreigners that correspond to goods and services produced in the domestic country using the factor inputs supplied by foreigners.

GDP is a better measure of the state of production in the short term. GNP is better when analysing sources and uses of income

Real and nominal values

Nominal GNP measures the value of output during a given year using the prices prevailing during that year. Over time, the general level of prices rises due to inflation, leading to an increase in nominal GNP even if the volume of goods and services produced is unchanged.

The national income of any country is important because it helps to assess the performance of a country over a period of time, usually in a year.

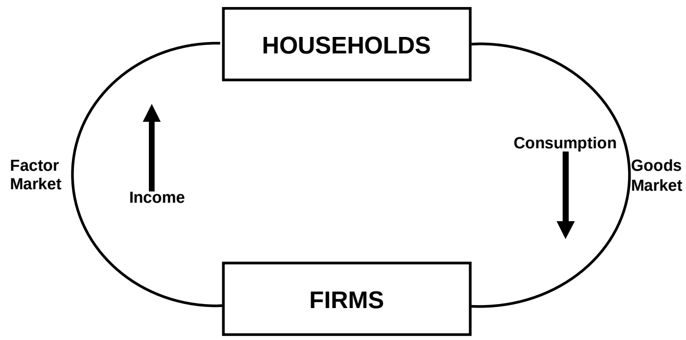
National income accounting is much like the accounting carried out by the individual firms to detect growth or decline in the profitability of a company. The national income figure that is calculated is used to compare the performance of the country in the previous years as well as the performance of that country with other countries.

In theory, the three methods of measuring the national income should provide the same figure as illustrated by the circular flow of income.

1.1 The circular flow of income

The circular flow of income describes how money moves between the different sectors in the economy. The expenditure of one sector is the income of another sector.

For a closed simple economy where there are only two sectors, firms and households, firms employ labour to produce goods and services. Firms spend on labour since households receive income for providing labour. Household income is spent on goods and services from firms.



1.2 THE NATIONAL OUTPUT OR THE VALUE ADDED METHOD

This is the total of consumer goods and services investment goods (including additions to stocks) produced by the country during the year. The production of goods and services in different sectors of the economy is added together. For example what is produced in the agriculture and fisheries, forestry, manufacturing, hotels, banking, national defence, education, health sectors etc, is all added together to arrive at the national income using the output method.
It can be measured by

- Totaling the value of the final goods and services and produced or
- Totaling the value added to the goods and services by each firm, including the government. This is done to avoid double counting and in order to use this method, a detailed knowledge of pricing is required. This explains why the output method is also known as **the value added method.**

The **usefulness** of this method is that it shows the changing shares of the industrial sectors in an economy, that is a sector which is expanding or falling it its contributions to the national income.

1.3 NATIONAL INCOME

Using this approach, the total factor incomes received by persons and firms for the provision of factors of production, is added. Income from employment, trading profits, rent and interest are all added together to arrive at the national income using the income method.

When using this method:

- Transfer payments or transfer incomes are excluded because the people receiving them do not produce anything. It includes private money gifts, sale of second hand goods such as a house, a car etc.
- Stock appreciation is deducted from the total because when inflation makes existing unsold stocks more valuable, production has not increased.
- Residual error (statistical discrepancy) is added to make statistics from each method balance.

1.4 NATIONAL EXPENDITURE

This involves adding together all total amounts spent on final goods and services by households, central and local government, including what is spent by firms on the net additions to capital goods and stocks in the course of the year.

The calculation of the national income using the expenditure method is what is known as the **aggregate demand**. This total spending is made up of consumption expenditure, plus investment expenditure, plus government expenditure, plus net exports (that is exports minus imports).

Adjustments have to be made for taxes and subsidies as they distort market prices, the idea is to measure the national expenditure, which corresponds to the cost of the factors of production used in producing the national product, which is known as national expenditure at factor cost.

The **usefulness** of this method is in detecting changing trends in consumption and investment, although this is the most widely used method, it trends to over estimate national output.

GROSS DOMESTIC PRODUCT (GDP)

The GDP is the first value arrived at in the national income calculations, before any adjustments are made. This is often referred to as the value of the output produced in the country during one year and if it increases in real terms, then it is a sign that the economy has grown. The GDP is calculated at **market prices,** but after taxes are deducted and subsidies are added, the GDP is at **factor cost**.

GROSS NATIONAL PRODUCT (GNP)

This refers to the value of the output produced by residents of a country in a year. It is arrived at after including the output produced by companies and individuals of a country but they are based abroad. In addition, output produced by foreigners and overseas companies in that country is deducted. This is summarized as net property income from abroad, which maybe positive or negative.

CAPITAL CONSUMPTION

Capital assets suffer wear and tear as such depreciation, termed capital consumption is deducted from GNP to arrive at the Net National Product or the National Income is short.

In summary

GDP at market Prices
- indirect taxes
+ Subsidies

= GDP at Factor cost
+ Net property income from abroad

= GNP
- Depreciation

= N I

1.5 ZAMBIA'S GROSS DOMESTIC PRODUCT BY KIND OF ECONOMIC ACTIVITY AT CURRENT PRICES (K'BILLION), 2003 – 2005 USING THE VALUE ADDED (OUTPUT) METHOD

KIND OF ECONOMIC ACTIVITY	2003	2004	2005*
Agriculture, Forestry and Fishing	4,244.6	5,568.2	6,856.6
Agriculture	1,008.2	1,249.5	1,526.0
Forestry	2,960.3	3,998.5	4,953.6
Fishing	276.1	320.2	377.0
Mining and Quarrying	564.8	809.6	980.5
Metal Mining	558.2	798.3	960.4
Other Mining and Quarrying	6.6	11.3	20.0
PRIMARY SECTOR	4,809.4	6,377.7	7,837.0
Manufacturing	2,241.0	2,827.7	3,458.1
Food, Beverages and Tobacco	1,397.2	1,726.6	2,145.5
Textile, and Leather Industries	352.9	450.7	491.2
Wood and Wood Products	164.7	222.2	283.7
Paper and Paper products	93.1	123.6	161.0
Chemicals, rubber and plastic products	178.9	231.7	286.3
Non-metallic mineral products	30.0	41.0	51.6
Basic metal products	3.1	4.0	4.6
Fabricated metal products	21.0	27.7	34.2
Electricity, Gas and Water	595.1	694.7	922.7
Construction	1,590.0	2,402.1	3,689.8
SECONDARY SECTOR	4,426.1	5,924.5	8,070.6
Wholesale and Retail trade	3,873.8	4,843.7	6,079.7
Restaurants, Bars and Hotels	527.7	670.9	895.9

Transport, Storage and Communications	1,058.2	1,252.3	1,408.3
Rail Transport	89.5	100.8	99.9
Road Transport	393.9	464.0	546.7
Air Transport	152.7	203.0	246.7
Communications	422.1	484.6	515.0
Financial Intermediaries and Insurance	1,847.7	2,282.7	2,776.9
Real Estate and Business services	1,341.2	1,691.8	2,105.8
Community, Social and Personal Services	1,757.0	2,046.5	2,529.1
Public Administration and Defence	683.0	723.9	869.4
Education	688.6	867.7	1,127.1
Health	252.4	292.8	329.1
Recreation, Religious, Culture	26.4	28.8	36.1
Personal services	106.6	133.3	167.3
TERTIARY SECTOR	10,405.6	12,787.9	15,795.8
Less: FISIM	(1,061.8)	(1,311.8)	(1,595.8)
TOTAL GROSS VALUE ADDED	18,579.3	23,778.3	30,107.6
Taxes on Products	1,899.9	2,219.1	2,541.1
TOTAL GDP. AT MARKET PRICES	20,479.2	25,997.4	32,648.6
Growth Rates in GDP	25.97	29.95	25.58
Current GDP per Capita (Current Prices)	1,852,017.00	2,317,860.00	2,909,857.00

Source: Central Statistical Office

1.6 Zambia's GDP by expenditure method, in Kwacha (bn) at current prices

	1990	1991	1992	1993	1994	1995	1996	1997
Total consumption	95	200	574	1316	1980	2779	3608	4752
Government consumption	17	35	102	192	313	489	714	857
Private consumption	78	165	472	1124	1667	2290	2894	3895
Total investment	20	24	68	223	284	394	582	701
Gross fixed capital formation	19	25	65	217	276	385	566	681
Public fixed capital formation	8	13	23	50	235	198	239	278
Private fixed capital formation	11	12	42	167	41	187	327	403
Changes in stock	1	-1	3	6	9	9	16	23
Net exports	-1	-6	-72	-57	-23	-175	-246	-284
Exports of goods and services	41	76	210	498	785	1109	1344	1715
Exports of goods	38	70	195	454	714	1027	1200	1565

Imports of goods and services	-41	-81	-282	-555	-809	-1284	-1590	-2000
Imports of goods	-33	-62	-233	-465	-671	-1034	-1275	-1601
Total GDP	113	218	570	1482	2241	2998	3945	5169

Source: Internet

2.0 USES OF NATIONAL INCOME FIGURES

- The main use is to assess the performance of an economy over a year. It is used as an indicator of a growing or a contracting economy, **Economic growth** or lack of it is assessed using the national income figures.
- The figures are used to indicate the overall standard of living, especially after dividing by the total population in a country to calculate the **per capital income**.
- This enables comparisons to be made between different countries. To ascertain which are rich and which ones are poor countries.
- To assist the government in managing the economy, using **Keynesian demand management**.
- The trade or Economic cycles depend on the national income figures. The figures are also used to estimate future movements.

2.1 LIMITATIONS IN NATIONAL INCOME CALCULATIONS

1. There are differences in the accuracy of the figures. Different countries collect and invest in data collection differently.

2. Economic welfare affected by medical and educational facilities per head. There is need to know what proportion of the national income is spent on the provision of better social sector facilities and not on defence! As with all mathematical averages, per capita income data does not take into account how the GDP is distributed amongst the population. If the income is unevenly distributed, then increases in the GDP per capita may disproportionately benefit a small group of high income earners and have little impact on reducing poverty. If GDP per capita data is to be used then its distribution must also be taken into account.

3. Arbitrary definitions, for example when calculating the national income, only those goods and services that are paid for, are normally included. Do it yourself jobs, such as gardening, repairing one's own car, housework etc, are excluded, and their exclusion distort the national income figure. These unpaid services, which are normally provided by housewives, are included in the calculation of the national income when done by someone else. If an individual lives a in a house for which he pays rent to the landlord, this will be treated differently from owning a house for which he no longer pays rent

4. Incomplete information, which can be attributed to, the high levels of subsistence sector, barter and black economies that are more pronounced in developing countries.

5. Danger of double counting, for example, the cost of raw materials and that of finished goods should not both be counted, this difficult to avoid when using the output method of calculating the national income.

6. Using any monetary data, such as GDP per capita over time, must recognise that output and incomes measures can increase for many reasons other than the country producing more goods.

   It is an increase in goods and services that is necessary if poverty is to be alleviated or peoples' livings standards rose. Output and incomes measures may increase because the rate of inflation has simply increased the money value of goods and service produced rather than their real value. **Real** GDP per capital would be a better indicator, as this is a measure of the physical value of goods and services produced. Real GDP is equal to the nominal GDP adjusted for price changes, (minus inflation).

   The different rates of inflation and the constant variations in the exchange rate within and in different countries make comparisons difficult.

7. The national income measures the standard of living. This has to relate to the size of the population. Some countries have a high-income figure and a correspondingly high population.

8. Some countries have high national income figure but are paying a high penalty for living beyond their means and borrowing heavily.

9. It should be remembered that GDP only includes output that involves a financial transaction i.e. is marketable. A considerable amount of Zambia's agricultural output is produced on small-scale communal farms for subsistence purposes. It is currently estimated that only 25% of production on communally owned land involves monetary transactions. The rest is not included in any national income calculations. Likewise the output of the informal sector will not be included.

10. Increasing national income and growth may occur at the expense of the environment rapidly growing economies may result in negative externalities. An agricultural sector that increases productivity by intensive use of pesticides and fertilisers or deforestation. may reduce future land fertility and worsen the level of poverty for future generations

2.2 FACTORS DETERMINING A COUNTRY'S NATIONAL INCOME

Income is not evenly distributed, and the factors determining a country's national income can be classified as internal and external, the latter resulting from a country's relationships with the rest of the world.

The most important internal factors are

Original Natural Resources

Natural resources are nature-given, such as mineral deposits, sources of fuel and power, climate soil fertility, fisheries, navigable rivers, lakes that help communications etc. New techniques allow national resources to be exploited while the exhaustion of minerals resources reduces national income. Some countries are well endowed by nature, and if the resources were well managed, then the national income would be high.

Where a country's economy is predominantly agricultural, variations in weather may cause national income or output to fluctuate from year to year, this happens to be the problem with most developing countries.

The nature of the people, particularly of the labour force

This includes the **quantity** of the labour force, the higher the proportion of workers to the total population and the longer their working hours, the greater s the national income figure.

Another factor is the **quality** of the labour force, their health, nutrition, energy, inventiveness, judgment and ability to organize them to cooperate in production, the climate, working conditions, peace of mind as well as education and training.

Capital Equipment

Productivity or labour will be increased if the quality of the other factors is high, for example, the more fertile the land, the greater is the output per man.

In addition, the quality of the capital equipment employed is the most important factor, the output of workers varies almost in direct proportion to the capital equipment, and the single most important material progress is investment in capital.

Consider the output per man where the majority of the farmers are using a hoe and an axe, while in advanced countries, farmers use tractors and combine harvesters!

Knowledge of techniques

This is acquired through the development of Research and inventions. The government can encourage this by financing research schemes. Alternatively the government can go into partnership with the private sector or offer incentives such as tax rebates to companies that are spending a lot on research and development. New inventions can bring in more income into the economy.

The organization of resources

One of the known factors that can improve production and therefore national income in most of the developing countries is the organization and the management of resources.

The leaders of any economy should have a vision for their countries. They have to be focused, set goals and objectives, have the right people and the right resources in order to achieve those objectives.

Political stability

A country has to be politically stable in order to produce. If the resources are being used on warfare, very little production of goods and services takes place. This again is a common problem in developing countries. Even if some are well endowed, they are not politically stable and the organization of resources is poor.

<u>The external factors are</u>

Foreign loans and investments

These are an injection of funds that lead to an addition of stock, adding to the national income.

Related to the above, **gifts** or **handouts** from abroad for the purposes of Economic development and defence improve the national income of the receiving countries.

Terms of trade

This is the rate at which one country's exports exchange with another country's imports**.** The terms of trade is not constant, it changes as export and import prices change.

Developing countries generally deal in the primary sectors and not in the secondary sectors in production. They export goods at low prices in their raw form, but import goods at relatively high prices as these are finished goods.

3.0 CHAPTER SUMMARY

The national income of any country is simply the total value of goods and services its people produce during the year. The national income can be measured in three different ways, the value added (output) method, the income method and the expenditure method. In theory, all the three methods should provide the same national income figure, based on the circular flow of income. A simple model of the circular flow of income assumes a two-sector economy of firms and households.

Factors of production move from households to firms, for the production of goods and services. Firms pay factor incomes, such as wages and salaries to households in exchange for the factors. The income earned by households is spent on goods and services produced by firms.

Calculation of the national income is very important in every economy as the figure has a number of uses. It is used to assess the performance of the economy over the years, to indicate the overall standard of living, and to enable comparisons to be made, from one year to the next, as well as comparisons between countries.

Unfortunately, there are a number of difficulties that are encountered in measuring national income, which provides unreliable testimony as to how the real welfare of the people has changed, and when making comparisons.

The national income, and therefore Economic growth depends on the natural resources of a country, the quality of the labour force and its participation rate, the capital equipment being used etc.

CHAPTER 2

NATIONAL INCOME DETERMINATION

After studying this chapter, the students should be able to:

- Explain the relationship between national income, consumption expenditure, savings and investment
- Discuss the factors that determine consumption, savings and investment
- Explain how the multiplier and the accelerator works
- Explain what the circular flow of income for an open (complex) economy is
- Discuss Economic or business cycles and their characteristic features

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1.0 Introduction

In macroeconomics, there are mainly theories of two schools of thought. The monetarists, the famous one being Milton Friedman, their arguments are mostly on the money supply and the effects of changes in the money supply.

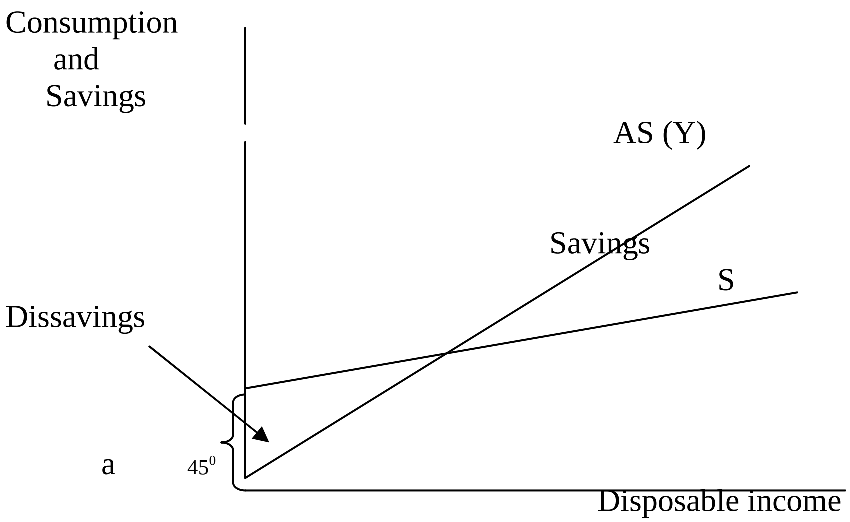
The Keynesians, advocates of Sir John Maynard Keynes. Keynes wrote a book entitled ***General Theory of Employment, Interest and Money***, published in 1936. His work was at the time of the great depression.

1.1 CONSUMPTION EXPENDITURE

From the circular flow of income, spending by households is termed consumption expenditure. It is an endogenous part of the circular flow of income.

Consumption expenditure depends on an individual's income, it is therefore a function of income.
C = F (Y).

As Y increases the level of consumption also increases but Lord Keynes maintained that each successive increment to real income is marched by a smaller increment to consumption expenditure, the rest is saved.



The extent to which consumption changes with income is termed the marginal propensity to consume (MPC). The marginal propensity to consume is the proportion of each extra kwacha of disposable income spent by households. That proportion of each extra kwacha of disposable income not spent by households is known as the marginal propensity to save (MPS).

If out of the extra kwacha increase, eighty ngwee is consumed, then the marginal propensity to consume is 80% or 0.8, and the marginal propensity to save is 20% or 0.2.

Therefore the MPC is the ratio at which the extra income earned is consumed, and it is denoted by the formula:

$MPC = \frac{\Delta C}{\Delta Y}$, and it depends on the slope of C. the steeper the slope, the larger is MPC

MPC depends on the slope of the consumption function, the steeper the slope, and the larger the MPC, which implies a small MPS.

The nature of the relationship between consumption and income is given by the straight-line equation:

$$C = a + by$$

Where
C = consumption
a = C when an individual is not working and income from employment is zero, it is also known as autonomous consumption.

b = MPC
y = income,

For example, assume K650 000 is required for a family of three people to survive, whether the head of the family is working or not. The K650 000 has to be found for the family to stay alive; it can come from social security, dissavings, begging, borrowing etc.

When in employment, for every extra kwacha earned, 80 ngwee is consumed. The equation becomes
C = 650,000 + 0.8Y.

Factors influencing consumption are income, interest rates, government policy such as taxation, which reduces the disposable income, hire purchase and other credit facilities, and invention of new consumer goods, which are later, introduced on the market.

Note that in any economy households, firms and government undertake consumption of goods and services

2.0 SAVINGS

Savings is defined as the part of income not spent, it is a withdrawal or **a leakage** from the circulation flow of income.

$Y = C + S$

$\therefore S = Y - C$

Therefore consumption and savings are two sides of the same coin and the consumption function tells us not only how much households consume, but also how they save. The factors that influence consumption naturally affect savings.

MPS is denoted by the formula = $\frac{\Delta S}{\Delta Y}$ where Δ is change.

Since any increment in income must be either spent or saved, MPC + MPS = 1

3.0 INVESTMENT EXPENDITURE

Investment is spent on the production of capital goods (houses, factories, machinery, etc) or on net additions to stocks such as raw materials, consumer goods in shops, etc.

In national income analysis, investment takes place only when there is an actual net addition to capital goods or stocks.

Investment is a major injection into the circular flow of income and affects national income and aggregate demand. Investment through the multiplier is needed to achieve Economic recovery. Investment is very dynamic, it determines future shape and pattern of Economic recovery.

Economic growth determined by technological progress, increase in size and quality of labour and the rate at which capital stock is increased replaced. Addition to stock should be greater than stock depreciation.

Investment therefore determines long-term growth, and both the private and the public sector can carry it out. If it is undertaken by the private sector, the government is expected to provide an enabling environment by stimulating business confidence by providing a stable Economic climate. Setting and achieving macroeconomic targets like low levels of inflation, controlling the money supply and therefore controlling the interest rates and consumption. The government can offer tax concessions or finance research schemes, sometimes in conjunction with the private sector.

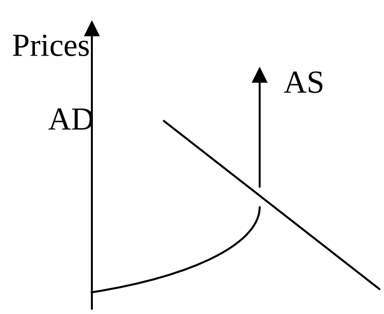
Keynesian demand management emphasizes the importance of the role, which the government plays to influence investment.

4.0 KEYNESIAN DEMAND MANAGEMENT

National income determination is a Keynesian concept. Keynes emphasized the importance of aggregate demand in the economy. The national economy could be managed by taking appropriate measures to influence aggregate demand up or down depending on whether there was a deflationary or an inflationary gap in the economy.

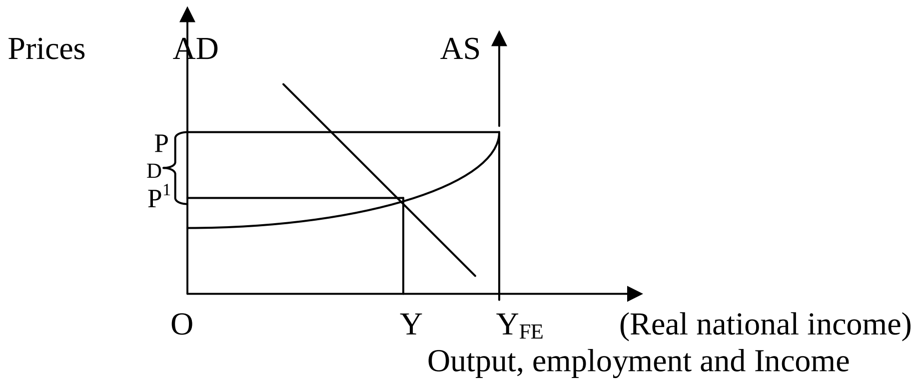
The aggregate demand (AD) is the total demand for goods and services in the economy. Aggregate demand is made up of consumption expenditure (C), government expenditure (G), investment expenditure (I) and exports (X) minus imports (M), that is AD = C + G + I + (X – M).

The aggregate supply curve (AS) is the total supply of goods and services in the economy, and a typical Keynesian aggregate supply curve is an inverse “L”. The explanation is that the AS curve becomes vertical when all the resources are fully employed.Keynes concentrated on shifting the AD, hence the name **Keynesian demand management**, and it involves manipulating national income by influencing C, I, G, or (X – M). According to Keynes, the equilibrium is where the AD is equal to AS at the full employment level. This is the ‘ideal’ position where all the resources are fully employed.



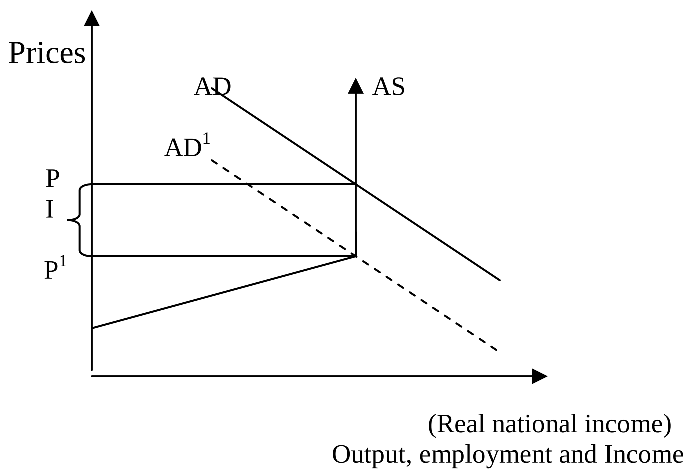
0 (Real national income)
Output, employment and Income

Suppose the economy is in a recessionary stage, and there is underemployment of resources, it means there is a deflationary gap.



Where D is the deflationary gap, this is the extent to which the government needs to increase AD to shift it to the right or upward to reach the ‘ideal’ full employment level in the economy.

Alternatively, the economy maybe experiencing inflationary pressures if AD is above the ‘ideal’ full employment position. The resources cannot be increased any further and this puts pressure on the prices of goods and services.



Where I is the inflationary gap, the extent to which the government needs to reduce AD to shift the curve from AD to AD^1, back to the equilibrium level.

4.1 THE MULTIPLIER PRINCIPLE

Keynesian demand management involves manipulating national income by influencing C, I, G, or X – M, while C is an endogenous part of circular flow of income, the others are **injections** into the circular flow. Any injection into the circular flow of income of a country starts a snowball effect.

If the government decides to build a big hospital in Zambezi district costing K10 billion, the increase in government expenditure through the construction of the hospital provides incomes to the factors of production employed in the construction of the hospital. Part of the K10 billion goes to the contractor as profits, part of it goes to the workers as wages and part of it is used for the purchase of building materials. The three groups who will earn the income will spend it.

Any expenditure becomes someone else's income, which is then in turn spent, generating a whole series of rounds of additional spending and income generation, the snowball effect. However, not all the income earned is consumed, some of it is saved. Savings is a leakage from the circular flow, other leakages from the circular flow of income are imports and taxes. The total amount leaked out is known as the marginal rate of leakages.

The effect on total national income of a unit change any of the injections into the circular flow income can be measured, it is called the multiplier.

$$\text{The investment, government or export multiplier} = \frac{\text{Eventual change in NI}}{\text{Initial change in I, G spending or X}}$$

The multiplier is denoted by the symbol K, and can be re-written as

$$K = \frac{\text{Total increase in NI}}{\text{Initial increase in NI}}$$

The shortcut method is to take into account the leakages, therefore

$$K = \frac{1}{\text{Marginal rate of leakages}}$$

Numerical example in a simple closed economy starts with the assumption that income is either consumed or saved. Suppose the MPC in the example above where there is an injection of K10 billion is 80% (0.8). & income increases by £200

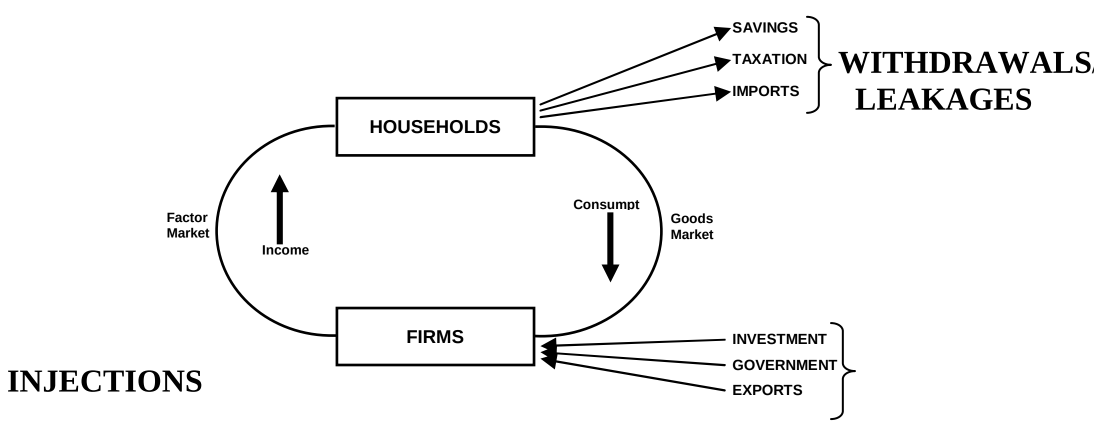
Stage		Increase in Expenditure (K'billion)	Increase in Savings (K'billion)
1	Income increase	10	-
2	80% consumed	8	2
3	A further 80% is consumed	6.4	1.6
4	A further 80% is consumed, etc	5.12	1.28

It works out to be $K = \frac{1}{1 - MPC} = \frac{1}{MPS} = \frac{1}{0.2} =$ 5 times

This translates to 5 times the initial investment of K10 billion, meaning that the eventual change in national income is K50 billion! If the marginal propensity to consume were much higher than 80%, then the multiplier effects would be much higher and vice versa.

Multiplier in a complex, open economy would be lower because all the leakages or withdrawals from the circular flow of income would be taken into account.

THE CIRCULAR FLOW OF INCOME FOR AN OPEN (COMPLEX) ECONOMY



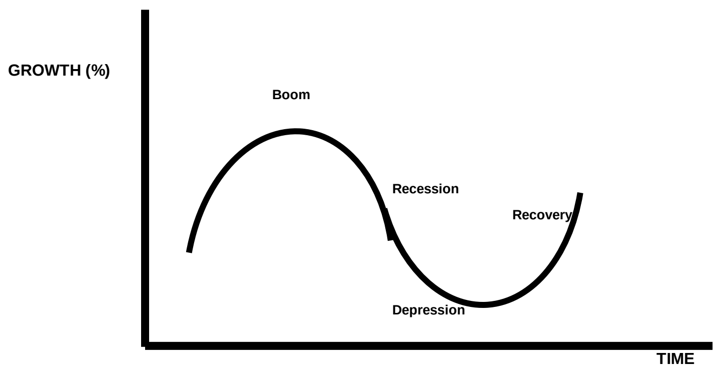
4.2 Accelerator theory

The accelerator relates to a small change in the output of consumer goods, which is said to result in a greater change in the output of capital equipment. This change in the production of capital equipment depends on the capital-output ratio.

The accelerator theory suggests that the level of net investment will be determined by the rate of change of national income. If national income is growing at an increasing rate then net investment will also grow, but when the rate of growth slows net investment will fall. There will then be an interaction between the **multiplier** and the accelerator that may cause larger fluctuations in the trade cycle.

In the multiplier principle, an increase in investment affects income and consumption, while under the accelerator, consumption affects investment. When the economy is expanding, and income as well as consumption is high, then the business sector is encouraged to produce more goods. Thus investment increases. The increase in investment leads to an increase in income and consumption, and so on.

The combined effect of both the multiplier and the accelerator results in the sequence of rapid growth in the national income followed by a slow growth, the business or trade cycles. These are made up of four phases namely, the recession, depression, recovery and boom.



When aggregate demand falls, businesses are discouraged, and both employment and production fall this is the recession stage. If this continues, then a full depression sets in.While a recession is quicker, recovery is slower because of lack of business confidence. Once recovery starts it is likely to quicken as business confidence returns. As output, employment and income increase, there is even more investment because of business expectations until a 'business boom' is reached.

4.3 THE PARADOX OF THRIFT

In theory, investment depends on savings. In order to increase savings, consumption must reduce because income is either consumed or saved.

Unfortunately, a reduction in consumption reduces business expectations, and the business sector reduces investment. A reduction in investment causes greater reductions in income. When income reduces, savings also reduces.

The paradox of thrift explains the working of the 'demultiplier'.

5.0 CHAPTER SUMMARY

The level of national income of any country is determined by the relationship between decisions by households to spend and save and decisions by the business community to invest.

The consumption function is C = a + bY. Where b is the marginal propensity to consume, this is the proportion of an increase in income, which is spent (consumed).

Therefore, the most important determinant of the level of consumption is income.

Savings is that amount of income not spent, and therefore, it also depends on income. However, there are other factors that influence consumption and savings, such as interest rates, inflation, levels of taxation, existence of financial institutions etc.

In theory, the national income remains the same from one period to the next if people decide to save an equal amount as the one, which business houses decide to invest.

Investment is the actual increase in stocks, the creating or buying capital equipment, not goods which are for immediate consumption. Investment depends mostly on the expected returns.
When national income is at its full employment level, and the total spending, commonly known as aggregate demand, is less than this figure, the deficiency is known as a deflationary gap.

If, on the other hand, at full employment, aggregate demand is in excess of the full employment national income, then an inflationary gap occurs.

An increase in aggregate demand gives rise to additional income due to the workings of the 'multiplier', but a reduction in aggregate demand gives rise to multiple reductions in income, called the de-multiplier.

The 'accelerator' theory links consumption expenditure to investment decisions. An increase in consumption expenditure results in more investments in capital goods in order to increase output to satisfy customers.

The combined effect of both the multiplier and the accelerator results in the business or trade cycles. A business cycle consists of simultaneous expansion in many fields of business activity, followed by widespread contraction.

The 'paradox of thrift' reflects the fact that a decision to increase the rate of savings may result in a decline in income.

Note the fact that the theory of income determination is developed on simplified models whose application in practice may be limited. However, Keynesian demand management is important because governments intervene in order to stabilize their national economies, and even if their interventions may not be fully successful, they do influence the level of Economic activity.

CHAPTER 3

MONEY AND INTEREST RATES

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After studying this chapter, the students should be able to:

- Define and differentiate forms of money
- Describe the basic characteristics of money
- Appreciate the functions of money in the economy
- Explain the demand for money
- Explain how the money supply is defined
- Explain how Interest rate is determined
- Discuss the Economic effects of interest rate changes

1.0 INTRODUCTION

Money is defined as anything that is generally acceptable in repayment of a debt. It is a medium of exchange, a legal tender.

The early forms of money where items that were in common use and generally acceptable, such as cattle, hides, furs, tea, salt, shells, cigarettes, etc.

These early forms of money had a lot of limitations, some items like cigarettes are not generally acceptable to be used by all to settle debts. Other early forms of money like cattle were not easy to carry around, and therefore not convenient. Some were perishable products like hides and as they were deteriorating with frequent handling.

In addition, there was no homogeneity in terms of size, colour and weight. The money that is used now such as the one, ten or fifty kwacha bank notes are similar and they are easily recognizable.,

Therefore, a good monetary medium must be:-

- Generally acceptable
- Fairly durable
- Capable of being divided into small units
- Easy to carry (portable)
- Should be relatively scarce
- Should be uniform in quality

1.1 The Origin of Money

Money came into use due to the abundance of some things, after producing in greater quantities than what is required for immediate consumption. There was need to exchange the surplus with another person for some other commodity. Initially, it was mostly the exchange of goods for goods, the barter system.

This method had a lot of limitations and inconveniences such as the 'double coincidence of wants'. Finding a person who had an item or service you wanted and in return you also have an item, which is wanted by that person, before any, exchange can take place.

There was also the problem of the rate of exchange. That is, how much of item X has to be given for item Y. Related to this was issue of one party to a transaction having only a large commodity to offer, but requires a small item in exchange.

The barter system is still being practiced among some people but to a very small extent.

Given the shortcomings of the barter system it is easy to appreciate and understand the functions that money performs in modern economies.

1.2 Functions of money

Money performs four main functions:

- **A Medium of Exchange**

This is its earliest function and the most important one. Money facilitates the exchange of goods, buyers and sellers meet and trade easily without the inconveniences of the barter system. This in turn promotes specialization, productivity, efficiency and wealth creation. Money is considered to be the 'oil, which allows the machinery of modern buying and selling to run smoothly'.

- **A Unit of Account and a Measure of Value**

Another drawback of the barter system is the difficulty of determining a rate of exchange between different kinds of goods especially large indivisible articles. Money acts as a common measure or standard value of the unit account of goods and services.

The value of goods and services is reduced to a single unit of account, and therefore the process of exchange is greatly simplified.

Money makes possible the operation of a price system and automatically providing the basis of keeping accounting records, calculating the profit and loss accounts the balance sheet etc.

- **A Store of Value**

Money is the most convenient way of keeping any income, which is surplus to immediate requirement. It makes possible a build up of stores of many things for future use, a store of wealth.
Money can also be stored in the form of other assets such as houses, but money is a preferred because it is a liquid store of wealth. This means that money can be converted almost immediately into a medium of exchange without ‘loss of face value’. Assuming that money is stable in value!

- **A standard for Deferred Payments**

Use of money makes it possible for payments to be deferred from the present to some future date. Borrowing and lending are greatly simplified, loans are taken and repaid in the form of money. Credit transactions cannot easily be carried out unless money is used. Given the assumed stability in its value, future contracts are fixed.

Credit transactions cannot easily be carried out unless money is used.

1.3 The Demand For Money

Demand for money means the desire to hold money, as distinct from investing it. This desire to hold liquid reserves is known as liquidity preference. According to Lord Keynes there are three motives for holding money.

- **The transactions motive**

Both consumers and businessmen hold money to facilitate current transactions. A certain amount of money is needed for every day requirements, the purchase of food, clothing, to pay casual workers etc.

- **The Precautionary Motive**

Most people like to keep money in reserve, in case an unexpected payment has to be made, e.g. illness, funeral, accident, car defects, household appliance defects, etc.

‘Active’ balances, depends any, fairly inelastic, less inelastic in the 2^{nd} and elastic in the 3^{rd} known as ‘idle’ balances.

- **The Speculative Motive**

This is for the purpose of accumulating more, since holding money in active balances does not yield any interest.

Speculation depends on the expectation of the future tend in securities e.g. attractive shares and govt. stock, this generally moves in the opposite direction with interest rates if interest rates go up i.e. people think the price of stocks will go down in the future they will hold money …

2.0 THE SUPPLY OF MONEY

Money supply in any economy is a very important part of government policy. Money supply has to be monitored as a guide to Economic policy.

The argument of the monetarists is based mostly on the money supply. However both the Keynesians and monetarists accept the importance of money supply.

The money supply is the stock of money existing at any particular point in time. It is basically made up of coins and notes in circulation as well as bank deposits. Coins and notes make up approximately only one fifth of the total money supply while bank deposits make up four fifth. This is because commercial banks 'create' money through the creation of credit and the creation of deposits.

2.1 MONETARY AGGREGATES

The definition of the money supply is carried out in order to measure monetary aggregates. In practice, money is measured either narrowly or broadly, with a very thin dividing line between the two.

- **Narrow Money**

This is money that is available to finance current spending, money that is held for transaction purposes, it highlights the function of money as a medium of exchange. Narrow money is designed in different ways, the narrow measure of money starts from MO (Pronounced as 'm nought'), is the narrowest definition of narrow money. It comprises mostly notes and coins in circulation, plus commercial banks operational deposits held by the bank of Zambia.

- **Broad Money**

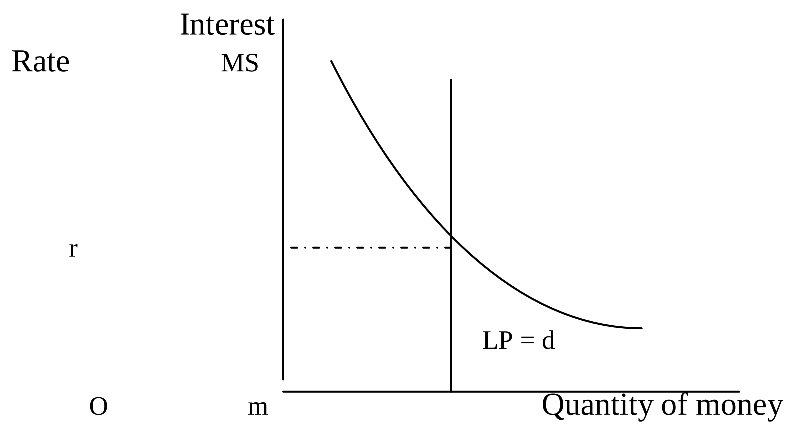
This is narrow money plus balances held as savings, that is the function of money as a medium of exchange and as a store of value. Therefore, broad money is money held for transactions purposes and money held as a form of saving.

Broad money includes assets, which are liquid but not as liquid as assets under narrow money. Broad money is also defined in different ways. The first broad definition of money is M4, and when foreign currency deposits are included, the definition is M3.

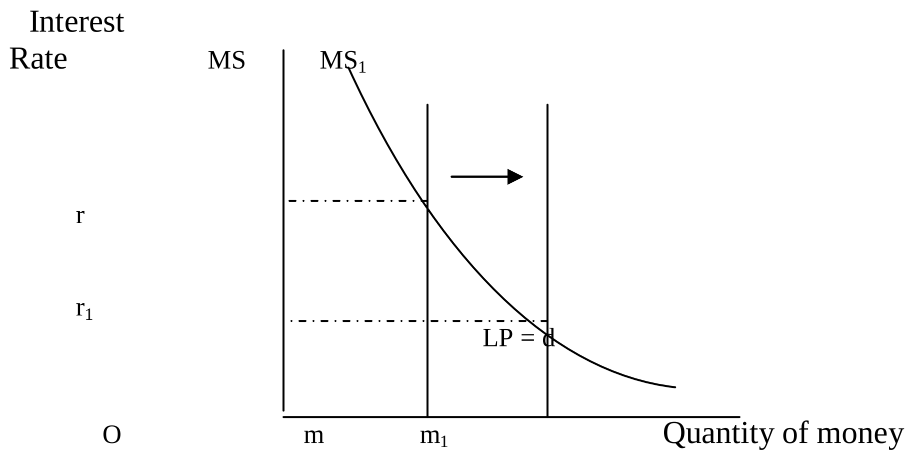
3.0 INTEREST RATE DETERMINATION

Interest is the price of money, and in the credit market, it is determined by the market conditions of demand for and supply of money. According to the Keynesians, the demand for money, liquidity preference is partly depend on the rate of interest. The supply of money is perfectly inelastic, the supply might increase or decrease depending on the government policy.

The equilibrium market rate of interest is determined at the point where the supply of money equals the demand for money



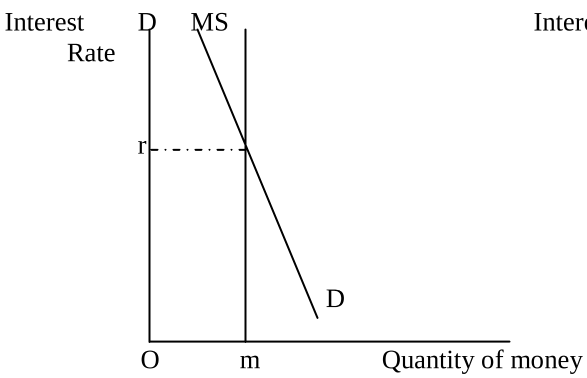
As in other markets, changes in either demand or supply conditions lead to a change in interest rates. In the Keynesian model, an increase in the money supply is associated with a fall in interest rates and vice versa.

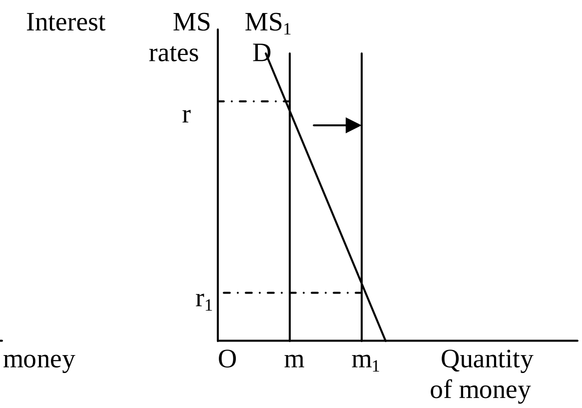


If the money supply increases from MS to MS_1, the equilibrium rate of interest reduces from or to Or_1.

3.1 THE MONETARISTS VIEW

Monetarists ensure that there are no three motives for holding money. Monetarists hold the view that money is held mostly for transactions purposes and enjoyment, and that demand for money is interest rate inelastic. As a result, any slight change in money supply leads to a big change in the rate of interest. This explains the need to maintain stability in the money supply in order to maintain stable interest rates.





Another argument by the monetarist is the microEconomic view known as the **loanable funds** theory (explained under **factor markets**).

3.2 EFFECTS OF INTEREST RATE CHANGES

Stable interest rates are important in any economy, if there is a large increase in the rate of interest, then the economy is affected in a number of ways.

- The cost of credit increases borrowing reduces and investment expenditure reduces.

- Spending by households also reduces savings is encouraged, since income is either consumed or saved, an increase in savings reduces consumption expenditure.

- Investment and consumption expenditure are components of aggregate demand, if spending by both households and firms reduces, then inflation is likely to be lowered. Low prices and less borrowing is a sign of less Economic activity.

- The foreign flow of funds increase financial speculators with ‘hot money’ are likely to be attracted to the high rates of interest.

- An increased flow of foreign funds puts pressure on the exchange rate. The high demand for the kwacha causes the kwacha to appreciate in value.

- A strong kwacha makes exports less attractive on the international market, reducing the demand for exports. Some workers are likely to be laid off, this reduces the level of Economic activity furthers.
- The business sector is also affected by the likely impact on profitability and investment projects that are appraised. High costs of borrowing compared to reduced cash flows due to a reduction in consumption expenditure.

3.3 VARIATIONS IN THE INTEREST RATES

Interest rates are determined by the market forces supply of and demand for money. In practice, there are several variations of interest rates that financial intermediaries apply, financial institutions do not give or charge exactly the same interest rates.

Finance bank, Indo-Zambia bank, Zambia National Commercial Bank etc all have their own rates that they offer to customers.

'Lending rates' given to surplus units (the depositors/savers who supply funds) and 'borrowing rates' charged to deficit units are different.

An individual bank can give or charge different rates to customers depending on estimated compensation for trying up the money, perceived 'risk' of the customer, amount and period of the loan etc.

In addition, there is the real rate of interest, which is the nominal rate of interest adjusted for **inflation**. The nominal rates of interest are the **expressed rates,** in monetary terms, hence they are also known as the money rate of interest.

The relationship between the inflation rates, the real rate of interest and the money rate of interest are:

(1 + real rate of interest) x (1 + inflation rate) = 1 + money rate of interest. This is usually approximated as real rate of interest + inflation rate = money rate of interest (nominal interest rate).

5.0 CHAPTER SUMMARY

Money is a medium of exchange, anything that is generally acceptable in the settlement of a debt. Early forms of money were items in common use, but had a number of limitations, hence a good monetary medium has a number of characteristics.

Money performs a number of functions in the economy, it is a medium of exchange, unit of account and measure of value, and it is a store of value and a standard for deferred payments.

The demand for money, according to Keynes, is the desire to hold money, and it is held as active balances, for the transactions and precautions motive, this depends on an individual's income and it is interest rate inelastic. Money is also held as idle balances for speculative reasons, and this depend on the rate of interest.
The money supply in any economy is simply made up of notes coins and bank deposits; however, money supply is a very important part of government policy, and it can be measured both narrowly and broadly.

Generally, the market forces of supply and demand determine interest rates. Interest rates should have some degree of stability, as they are very important in Economics and in the business environment.

In practice, there are variations in the rate of interest, which affect savings and loan repayments.

CHAPTER 4

FINANCIAL SYSTEMS AND MONETARY POLICY

After studying this chapter, the students should be able to:

- Appreciate the flow of funds and financial intermediation
- Identify the main elements of the monetary and financial system
- Explain the importance of the monetary environment to the business sector
- Explain the functions performed by commercial banks and the central bank.
- Understand how commercial banks ‘create’ money
- Explain the Economic role of government through monetary policy and its basic instruments
- Understand commercial bank’s conflicting objectives of liquidity, profitability and security

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1.0 INTRODUCTION

A financial system is made up of financial markets and financial institutions, it may also include formal and unregulated systems of finance, such as moneylenders, trade credit, micro finance and co-operative credit.

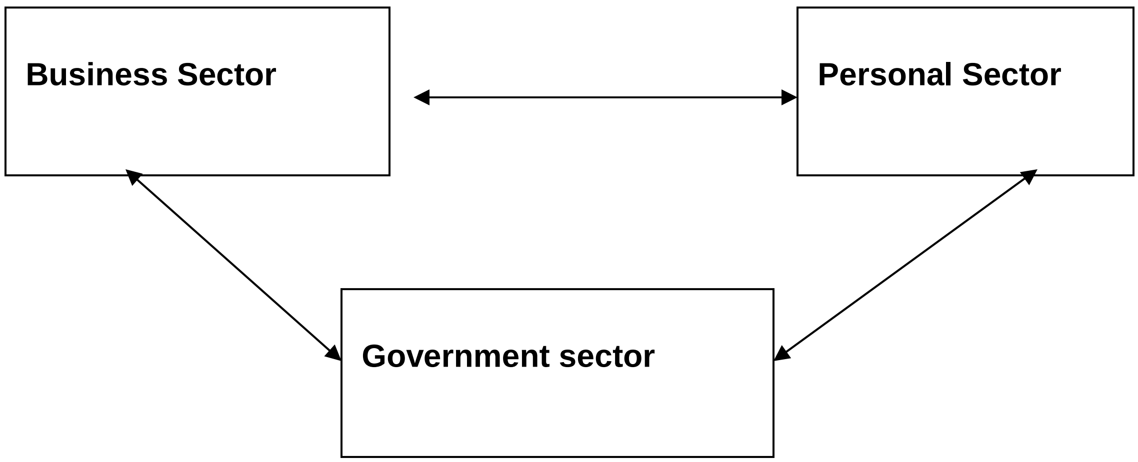
Financial markets are the capital market, the money market and the foreign exchange market. Financial institutions are commercial banks, building societies, insurance companies, national savings and credit bank etc.

A financial system brings in many benefits in the economy, it facilitates payments, raise the level of savings and investment. Capital is accumulated and allocated to uses where there are highest returns. The system helps to provide a means of transferring and distributing risk across the economy. Risk is diversified or pooled among a large number of savers and investors, by offering assets with different degrees of risk, financial institutions assess and manage risks and assign to individuals having different attitudes and perceptions towards flow of funds.

1.1 THE FLOW OF FUNDS

Funds flow between three sectors in the economy. Individuals give and lend money to each other. Organizations lend money and purchase goods from each other. The central government provides funds to the local government and loss making nationalized industries.

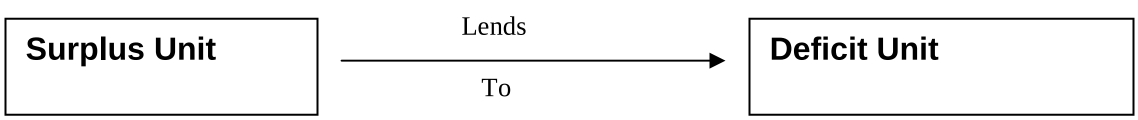
In addition, there is also a flow of funds between the different sectors of the economy.



A go-between, known as a financial intermediary, such as a commercial bank or building society, facilitates the flow of funds. Banks provide a means by which funds can be transferred from surplus units in the economy to deficit units.
Linking lenders to borrowers.

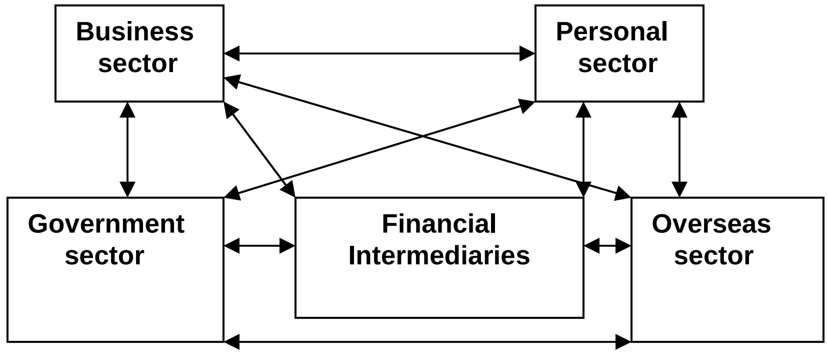
An individual deposits saving with a bank, the bank provides a loan to a company, as explained under credit creation.

If no financial intermediation takes place, lending and borrowing is direct.



Direct lending is also known as financial disintermediation. Savers wishing to lend have to find a trust worthy borrower. The lender bears the risk of default.
The borrower has to locate savers with money to lend. This results in high information cost and high default risk.

In practice, financial intermediaries also lend abroad, as well as borrow from abroad. Therefore, a detailed diagram of the flow of funds showing the role of financial intermediation in an open economy is as shown below:



1.2 BENEFITS OF FINANCIAL INTERMEDIATION

- A convenient way in which lenders save money
- They can package the amounts lent by savers and lend on to borrowers in bigger amounts (**aggregation**)
- They provide a ready source of funds for borrowers.
- The lenders, capital is secure, bad debts are borne by the financial intermediary.
- They bridge the gap between the wish of most lenders for liquidity and the desire of most borrowers for loans over longer periods. This is known as **maturity transformation**.
- They provide tangible returns to savings i.e. Interest rates.
- They act as an important medium for the implementation of financial (monetary) policies.

1.3 FINANCIAL INTERMEDIARIES

The banking system is made up of commercial banks and the central bank, and the non-banking financial intermediaries e.g.

- Building societies
- National credit and savings bank
- Insurance companies invest premiums paid on insurance policies by policyholders.
- Pension funds are mobilized through forced contributions from employees, and these are invested in financial markets and real estate.
- Investment trust companies invest in the stocks and shares of other companies and the government. Investment trusts simply trade in investment.
- Unit trusts are similar to investment trusts in that they invest in stocks and shares of other companies, but they enable small investors to invest with minimum risk. Unit trusts comprise of portfolios (stocks or shares in a range of companies or for example shares in mining companies), the trust creates a large number of small units, which are then sold to individual investors.

- Development banks are specialized financial institutions that provide long term and medium term funds.
- **Venture capital** is investment in **long-term, risky equity finance**, where the reward is an eventual capital gain, a takeover, a trade sale or a stock market flotation, rather than dividend income or interest. The main providers are investors in industry.
- **Hire-purchase** companies, facilitate the acquisition of physical assets through **extension of credit.**

2.0 FINANCIAL MARKET

The market for finance is basically divided into the following

- Money markets
- Discount market

Short term and commonly known as the money market.

- Capital market
- Stock market

Long term capital market

Money markets deal with short-term finance i.e. lending and borrowing for less than one year. Active participants in the market are commercial banks, the central bank, big organization and brokers. Financial instruments traded are:

- Certificate of deposits
- Commercial paper, I Owe You (I O U)
- The discount market, which performs the following functions:
  - Help finance short-term trade debts by purchasing commercial bills at a discount.
  - Create a market in bills of exchange

The money markets also includes

- Inter-bank markets for unsecured loans between banks; this facilitates smoothing out fluctuations in receipts and payments of banks, and determines likely future trends in interest.
- Eurocurrency markets, which are short-term deposits and borrowing mainly for the purposes of working capital. It is in other currencies either than that of the denomination of a particular country. With the growth in mergers largely involving multinational companies, there is need to shop around for favourable terms and avoid domestic government credit restraints.

2.1.0 Capital Markets

This market is divided into primary market for the new issue of shares, and the secondary market, which deals with reselling of existing securities. Instruments traded on the capital market are equity shares, mortgages, corporate and government bonds etc.

Note that the international capital market for medium term and long term are known as the eurocredit (for working capital and investment purposes) and the Eurobonds respectively.
The Eurodollar or the Eurobond market deals with long-term finance. Bonds are issued by very large companies, banks, governments and institutions such as the European Union. The bonds are denominated in a currency other than that of the borrower.

The bonds are bought and traded by investment institutions, banks etc.

2.1.1 The Stock Exchange

The Lusaka stock exchange is an organized capital market, which plays an important role in the functioning of the economy.

- It makes it easy for large firms and the government to raise long term capital, by providing a market for borrowers and investors to come together
- It publishes the prices of quoted (or listed) securities, which are then reported in the media.
- It tries to enforce certain rules of conduct for its listed firms and for operators in the market to provide investor confidence, and make them more willing to put their money into stocks and shares.
- It is a primary market for newly issued shares. Note that a firm's new shares are issued by an issuing house with the help and advice of a stockbroker.
- A secondary market exists for the buying and selling of existing shares, the buyers of new issues know that they can sell them in future.

The capital instruments traded are equities/securities such as ordinary shares, preference shares and debentures, as well as government bonds/gilds.

It also acts as an Alternative Investment Market (AIM), where small companies gain access to capital, under less stringent, less costly procedures.

A stock market is usually given as an example of a perfectly competitive market, even if it is affected by political factors such as wars, elections or any other form of uncertainty, including the general mood of the business.

3.0 COMMERCIAL BANKS AND CREATION OF MONEY

Commercial banks are financial intermediaries, with the same roles and benefits as other financial institutions.

A Commercial bank's activities can be one or more of the following:

- Clearing – settling payments
- Retail – traditional banking, offers small deposits and small loans to customers.
- Wholesale – bank dealing with large quantities

- Investment-also known as merchant banks, are specialized and deal with corporate customers.

3.1 Functions of commercial banks

Banks provide a payment mechanism and a place to store surplus money; they also provide means of obtaining and selling foreign exchange.

Commercial banks advise and assist companies in the issue of shares, and give investment unit trust advice and business.

They also engage in financial leasing, debt factoring or collection management, including executorships (trustee) services.

Banks finance import and export operations and investments.

The most profitable business of commercial banks is lending money in the form of overdrafts, discounting bills of exchange and loan facilities. This particular function is always expanding because banks create credit, create deposit and therefore create money.

3.2 Credit creation

Commercial banks are financial intermediaries, who accept deposits and extend loans. They operate on the assumption that they know from experience that customers only withdraw a fraction of what they have deposited and that not all depositors withdraw all their cash at the same time. Therefore, banks only keep a small fraction of their assets as actual cash, known as **fractional reserve** system.

To simplify the explanation on credit creation, assume that there is only one commercial bank, which is already in business with lenders and borrowers. Assume also that the bank knows from experience that the reserve ratio, that is, funds that a commercial bank must keep, as actual cash is 10% of a customer's deposit.

If a customer deposits K100 million in the bank, then the bank's balance sheet on day one with appear as follows:-

DAY ONE

Liabilities			**Assets**
	Km		**Km**
Share Capital	10	Fixed assets	10
Customer deposit	100	Cash	100
	110		**110**

Note that the share capital is used to finance fixed assets, and therefore, can be ignored; it is not part of the credit creation process.

If the bank loans 90% of the deposit, then the bank's balance sheet will appear as follows:

DAY TWO

Liabilities		***Assets***	
	Km		Km
Share Capital	10	Fixed Assets	10
Customer deposit	100	Cash	10
		Loan	90
	110		110

An individual, firm or a government borrows money for a purpose. Suppose the loan of K90m is used to purchase a motor vehicle, the person receiving the money deposits it in the bank, and the bank's balance sheet will appear as follows:

DAY THREE

Liabilities		***Assets***	
	Km		Km
Share Capital	10	Fixed Assets	10
Deposit - original	100	Cash	10
New deposit	90	Cash from new deposit	90
		Loan	90
	200		200

The bank will again maintain only 10% of the new deposit as actual cash and lend the rest to customers. This process continues, however, the credits and deposits being created reduce until it becomes too small to generate a fresh loan.

It is possible to calculate the total deposits created from any initial deposit and to come up with the final balance sheet by using the formula.

$$D = \frac{C}{R}$$

where D = final deposits created
C = initial cash deposit
R = cash reserve ratio

$$D = \frac{100}{10\%} = 100 \times \frac{100}{10} = 1\ 000$$

From the initial deposit of K100m, the bank's final balance sheet appears as:

Liabilities	Km	Assets	Km
Share Capital	10	Fixed Assets	10
Total deposit	1000	Cash	100
		Loans	900
	1010		1010

Note that the bank has created credit worth K900m from the initial deposit of K100m, this has resulted in additional deposits of K900 which is in circulation as part of the money supply.

In practice there are several banks in any economy and billions of money is 'created', but the process remains basically the same. This process is very important to know because it is closely linked with the money supply in the economy and as such the level of Economic activity.

3.3 LIMITS TO CREDIT CREATION

A deposit of K100m with a cash credit multiplier of 10% may not result in K1 billion total deposits. Some of the restrictions on credit creation may be summarized as:

- Since there are several banks, an individual bank cannot adopt an expansionist policy, unless other banks are willing to do the same. The clearing bank may have an overcautious credit policy, and restrict lending.
- A bank's ability to lend depends on its ability to acquire deposits, this may sometimes be limited by the lack of public confidence in the banking sector, and their inability to make abnormal demands for cash.
- Lack of collateral security may also hinder the process of lending and borrowing.
- The government, through the central bank, may decide to restrict lending and borrowing, in order to control the money supply in the economy.
- There is an inverse relationship between the cash reserve ratio and the money that is 'created' by banks. In developing countries, the cash ratios are relatively high, this means less Economic activity
- If the loans are spent on imported goods, then the foreign banking system benefits instead of domestic banks.

4.0 CONFLICTING OBJECTIVES OF PROFITABILITY, LIQUIDITY AND SECURITY

A commercial bank's assets and liabilities reflect a balance between conflicting demands of liquidity, profitability and security. A commercial bank has to serve the interests of its shareholders, which is maximising profits, and to attain this, the bank has to lend as much as possible. However, the bank has to ensure that it has adequate liquidity to meet the cash demand from depositors. As far as depositors are concerned, their money is secure at the bank.

- **Liquidity**

Used to settle daily cash withdrawals from customers and to settle accounts with other commercial banks in the clearing system, but balances for these purposes earn no interest and are unprofitable. Banks have to make sound investment policy, by investing in assets that can be easily converted into cash.

- **Profitability**

A commercial bank's profit is normally obtained from interest charged on assets minus interest paid on liabilities.

Commercial banks have an objective of trading profitably like other commercial organizations. To pursue this objective they need to earn high interest rates. Therefore, they have to lend for a long term and to high-risk customers. This reduces the choice of liquidity and security.

- **Security**

Commercial banks are expected to act prudently to safeguard the interests of depositors and shareholders; this however reduces opportunities for profitable lending.

5.0 THE CENTRAL BANK

This is the principal financial institution in a country, and it acts as a regulator of the banking system. Zambia's central bank is known as the Bank of Zambia (BOZ), it was established to take over from the Bank of Northern Rhodesia on the 7th of August 1964 although its Act was only passed in June 1965.

The central bank does not deal directly with the general public, but provides services to the commercial banks and the government and manages the money supply on behalf of the government and the people of Zambia for the good of the economy and not for profit maximization.

The Bank of Zambia's stated functions are:

- To ensure appropriate monetary policy formulation and implementation
- To act as the fiscal agent of the Government
- To license, regulate and supervise banks and financial service institutions registered under the Act to ensure a safe and sound financial system
- To manage the banking and currency operations of the Bank of Zambia ensuring the provision of an effective service to commercial banks, Government and other users.

5.1 Other functions of a central bank

- Issuing notes and coins. Bank of Zambia has the sole right to issue coins and notes in the economy.

- It acts as banker to the government. The government is the most important customer of the central bank. In addition, the central bank performs many tasks for and on behalf of the government such as:

  - Keeping the central government accounts and the accounts of the many other government departments.
  - Giving assistance by means of “ways and means” advances if the account is in “red” or overdrawn
  - Conducting government borrowing through the issue of Treasury Bills and government stock.
  - Advising the government on financial matters

- It acts as banker to the commercial banks, use the central banks use the central bank in the same way as private customers use the commercial banks. Commercial banks:

  - Draw coins and notes from their balances at the central bank as required.
  - Set off the net payments which has to be made to other banks as a result of the days clearing by drawing on the balance held at the central bank
  - Take advice on financial matters from the central bank

- Acting as lender of the last resort
- Holding the gold and foreign currency reserves in the exchange equalization account, which the central bank uses to stabilize or manage the kwacha.
- The central bank maintains close contact with other central banks and monetary authorities of other countries with the aim of achieving greater international monetary stability.
- It works in conjunction with international monetary organization like the international monetary fund and the world bank
- The central bank manages the national debt. It is responsible for floating new loans and the repayment of maturing loans plus interest as well as payment of interest to holders of government securities.

5.2 BANKING SUPERVISION

- **Capital Adequacy Rules**

Commercial banks make profits by charging interest on amounts borrowed, some amounts are not repaid, bad debts arise; hence the need to set capital adequacy ratios. The central bank imposes certain rules and requirements.

- **Liquidity**

Commercials banks need to hold money to meet customer demand, the central bank discusses with each individual commercial bank the adequacy of its stock of liquidity and can advise on changes.

- **Provision.**

The central bank encourages commercial banks to make adequate provision for bad and doubtful debts. In addition a bank is not allowed to lend more than 10% of its capital base to one single borrower.

- **Systems**

Each commercial bank reports periodically on its procedures and controls. The central Bank examines methods for monitoring credits risks, its systems for recoverability, its arrears patterns etc.

- **Personnel**

The directors, managers and large shareholders of banks have to satisfy the central bank that they are "fit and proper" people for such positions that is in terms of honesty, competency, diligence and are of sound judgment.

5.3 MONETARY POLICY

Monetary policy is becoming increasingly important in Economic management. Fiscal (budgetary) policy is once a year, in between, the government has to rely on monetary policy. The central bank controls the money supply in order to help the government achieve its macroeconomic objectives.

Monetary policy is decisions and actions of the government regarding the supply of money and its price (the rate of interest). An increase in the money supply, which is loose monetary policy leads to a lot of borrowing and spending. With too much money in circulation, inflation as well as an external trade deficit is the likely result.

To reduce the money supply, the central bank has to curtail the borrowing and spending by limiting the commercial bank's capacity to create credit, create deposits and therefore 'create money'. The instruments that the central bank uses to control the money supply are:

➢ **Open Marketing Operations**

By intervening in the open market to buy or sell securities, the central bank can directly influence the size of bankers' deposits. If the central bank wants to reduce the rate of inflation, it has to control (reduce) the money supply, and if it sold securities, e.g. treasury bills if it is a short term measure or government bonds if it is a long-term measure, the central bank receives payment by cheques drawn on commercial banks. This brings about a reduction in commercial banks deposits as well as the amount of money circulating in the economy.

➢ **Bank Rate (interest rate changes)**

The importance of central bank rate is that other rates of interest used, depend on it, the rate charged to discount houses, the rates charged on advances to customers and the rate offered on deposit accounts.

These rates move up or down with central bank rate. To check inflation, i.e. to reduce the money supply, the interest rate is raised to make credit expensive and as such discourage people from borrowing**.**

- **Special Deposits**

To reduce the cash basis for credit creation and to contract credit, the central bank can request commercial banks to place specified amount or to increase the percentage of these specified amounts, which are supposed to be kept in frozen accounts with the central bank. The government pays interest on the 'special deposits'. When following an expansionary policy, to encourage lending, the special deposits are returned.

- **Assets Ratios**

The central bank dictates or compels commercial banks to keep certain proportions of specified assets. To control inflation the ratio is raised.

- **Directives (moral suasion)**

This is a direct instruction from the central bank to the commercial banks to restrict their lending.

The directive can be in two forms:

- **Qualitative,** this is when commercial banks are requested to restrict lending only to purposes regarded as being in the national interest. Commercial banks would be encouraged to lend only to important sectors in the economy such as agriculture, mining and manufacturing.

- **Quantitative,** this is when banks are instructed to reduce their lending by a required amount.

Note that directives are easy to enforce in a command, planned Economic system. In a liberalized market Economic system, firms including banks have the freedom to meet new market demands without much government intervention.

5.4 Limits to Monetary Policy

In practice, monetary policy is not easy to achieve, not only because of a liberalized market Economic system, but because it also depends on commercial banks curtailing credit, given the fact that lending is the most profitable business of commercial banks, the banks find ways of circumventing the policies.

In addition, central banks face problems in applying monetary policy, for example

- A central bank may lack adequate, detailed, up to date information on the economy and the money supply.
- The central bank has to closely supervise the commercial banks to ensure that they have reduced their lending to customers, in order to reduce the money supply in the economy.

- Conflicting objectives of reducing the money supply, which results in an increase in the rate of interest, lower investments, less Economic activity and increased unemployment. The government has to trade inflation for unemployment or vice versa.
- The reluctance of the central bank to undermine initiative and commercial banks' ability to make profits, as mentioned earlier, lending is the most profitable business of commercial banks.

6.0 CHAPTER SUMMARY

Financial systems are made up of financial institutions and financial markets. Financial institutions enable the three sectors of the economy, the households, firms and government to borrow and lend to each other as financial intermediaries.

Some of the functions of financial intermediaries are to provide savings facilities and tangible returns to savers, maturity transformation, ready source of funds for borrowers, and as a medium for the implementation of monetary policy.

The financial market is composed of the money and the capital market, dealing in short and long-term finance respectively. Both markets are divided into primary and secondary markets, meaning issuance of new securities and trading in second hand securities respectively.

One of the significant groups of financial intermediaries is the banking sector. Commercial banks perform a number of functions, such as providing a payment mechanism, offering financial advice, financing import and export operations etc.

More importantly, the operation of the banking system increases the money supply, as commercial banks 'create' money. This depends on the cash deposited in the banking system and the cash reserve ratio.

When pursuing the most profitable business of commercial banks, which is lending to customers, a bank has to try and balance liquidity, profitability and security.

The central bank is the primary financial institution in any country, and it performs a number of functions. The most important of which is banking supervision and controlling the money supply in the economy, known as monetary policy.

CHAPTER 5

INFLATION AND UNEMPLOYMENT

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After studying this chapter, the students should be able to:

- Define and explain causes of inflation
- Understand how to measure changes in the value of money
- Identify the negative effects of inflation
- Explain the measures used to control inflation
- Define and explain types of unemployment
- Understand how to measure changes in unemployment levels
- Identify the negative effects of unemployment
- Explain the relationship between inflation and unemployment
- Appreciate the **supply-side policies**

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1.0 INTRODUCTION

In any economy, the government can follow expansionary or contractionary monetary or fiscal policies by either increasing the money supply and increasing aggregate demand or reducing the money supply and reducing the aggregate demand respectively. Unless the 'supply side policies' are put into effect, a government cannot easily control both inflation and unemployment at the same time. One Economic 'evil' has to be 'traded off' for the other.

2.0 INFLATION

Inflation is a sustained rise in the general price level of goods and services. It is measured using price indices.

Inflation can be classified between two extremes depending on the speed at which prices are changing. **Creeping inflation** is when there are small price increases while **hyperinflation** is the worst case of inflation. The prices of goods and services change very rapidly.

2.1 CAUSES OF INFLATION

There are three main causes of inflation, one view from the Monetarists, and two views from the Keynesians. Demand-pull and cost-push are essentially Keynesian explanations of inflation. Monetarists reject these and believe that inflation is caused by an increase in the money supply. Keynesians on their part do not accept that an increase in the money
Supply actually causes inflation.

They believe that an increase in the money supply is an indication that there is inflation in an Economy. It is not a cause of inflation.

2.2 MONETARISTS VIEW AND THE QUANTITY THEORY OF MONEY

Monetarists consider the increase in the money supply as the only cause of inflation. The argument of the monetarists is based on the quantity theory of money. The theory is summarized as the fisher equation, Irving Fisher developed the Fisher equation of exchange. It appears in various guises, the most common is:

MV = PT
where:

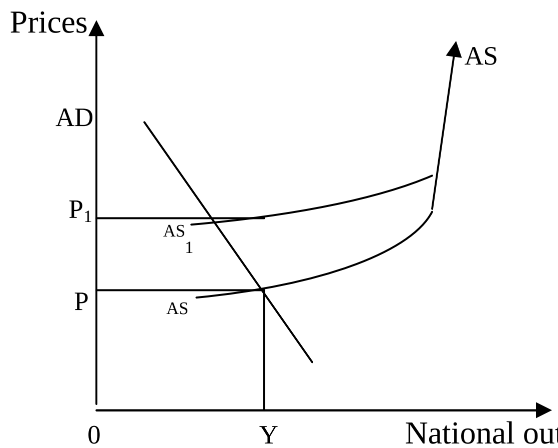
M is the amount of money in circulation
V is the velocity of circulation of that money
P is the average price level and
T is the number of transactions taking place

MV = PT states that money supply multiplied by the velocity of circulation equals the price level multiplied by total transactions. This equation is true by definition since receipts are equal to expenditure. **PT** can therefore be thought of as equivalent to **National Expenditure.**

Assuming that **V** is constant in the short run as it is determined by the money supply, and **T** is also fixed in the short run. Then, an increase in the money supply would lead to an increase in the general price level.

2.3 COST-PUSH INFLATION

This inflation is caused by an autonomous increase in the costs of production, considered as cost-push factors. These may then cause **cost-push inflation**. Cost-push factors may be changes in wages, changes in the exchange rate, which change the price of, imported raw materials or perhaps changes in indirect taxation. Cost-push inflation occurs when a company's costs rise and to compensate, a firm has to put prices up. Cost increases may happen because wages have gone up or because raw material prices have increased. The increase in the costs, with aggregated demand remaining uncharged, causes the aggregate supply curve to shift to the left from AS to AS_1, and price increases from OP to OP_1.



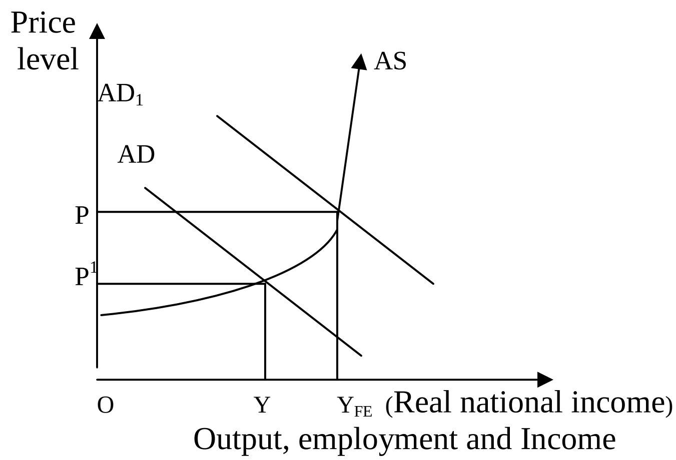
Cost-push factors that can contribute to the increase in the cost of production include:

- Strong, powerful trade viewers who force employers to concede high wage increases, costs then rise and these are later passed on to consumers in the form of price increases. This situation is worse during periods of low unemployment.
- Import cost inflation, especially for a country which depends on one or more of the
- Imported raw materials or other imported inputs
- Imported finished products, capital equipment and more especially the ever
- Increasing price of imported fuel.
- High indirect taxes such as when there is an increase in value added tax or excise duties, consumers simply notice an increase in prices.

2.4 DEMAND-PULL INFLATION

If there is an excess level of demand in the economy, this will tend to cause prices to rise. This type of inflation is called demand-pull inflation and is argued by Keynesians to be one of the main causes of inflation. Inflation occurs when increases in aggregate demand pull up prices, with aggregate supply remaining constant.

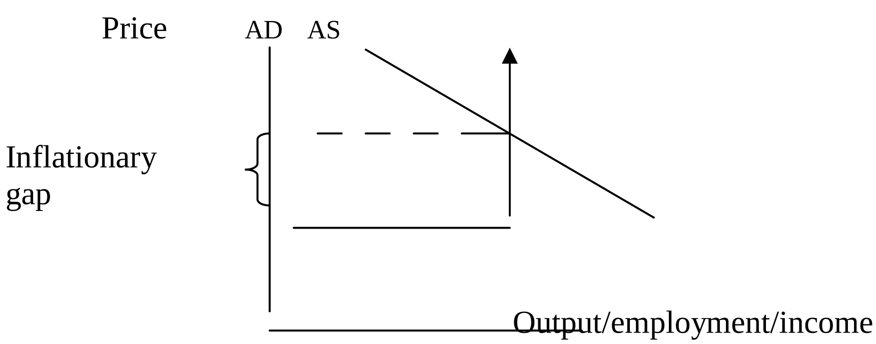
The aggregate demand is the total demand in an economy, made up of government expenditure, consumption expenditure, investment expenditure and exports minus imports. Any increase in one or more of these components of aggregate demand can put pressure on prices. As demand increases from AD to AD_1 there is increasing inflationary pressure on prices. This is demand-pull inflation - *"too much money chasing too few goods."*



Increase in demand can be caused by either expansionary monetary or fiscal policies. If there is a high public sector net cash requirement, then total demand in the economy is stimulated.

The Keynesian original aggregate supply curve is an inverse "L". According to them, the pressure on prices is when aggregate demand expands after the full employment of resources, before that point, an increase in aggregate demand acts as an incentive for firms to increase output.

When the resources are fully employed, the aggregate supply curve becomes vertical, and if aggregate demand increases beyond this point, an inflationary gap is created.



2.5 ANTI INFLATIONARY MEASURES

To control inflation, first, it is necessary to know the cause. Unfortunately, this is difficult to do because inflation tends to feed on itself and there is the price wage spiral.

Suppose the prevailing inflation is demand driven, then, measures to reduce aggregate demand should be put in place; such as tight fiscal and monetary policies like increasing direct taxes and interest rates to reduce consumption expenditure and investment expenditure respectively. Government expenditure should also be reduced. This means that the government must aim for a budget surplus, by increasing its income through increased taxes, but reduce government spending, the excess money should be kept frozen at the central bank.

If the inflation is due to an increase in the money supply then the government should attempt to reduce the money supply by reducing commercial bank lending using the instruments mentioned under the control of the money supply. These are open market operations, increasing interest rates and asset ratios to discourage lending. Directing commercial banks to reduce their lending and requesting them to make special deposits at the central bank.

If the source of inflation is an autonomous increase in the cost of production, then measures should be taken to stop the wage-price spiral, reducing the power of trade unions or match the increased costs with increased productivity.

A country's currency can be allowed to depreciate in order to discourage imports, while encouraging exports, this means increased production. An increase in supply lowers the price.

Prices and incomes policy that is wage and price controls can also be instituted to control inflation. This means freezing prices and incomes. This may not work well in a liberalized market economy. It also means controlling the consequence and not the cause of inflation!

2.6 ECONOMIC CONSEQUENCES OF INFLATION

A little inflation is considered to be good for any economy as it provides an impetus for firms to increase output. High prices are a sign that there is a high demand for goods and services and there is a prospect of higher profits.

Generally, the negative effects of inflation are as follows:

- Inflation redistributes income
- Retired people who are on fixed incomes suffer a lot from inflation. Some people earn K20, 000 per month as pension. At the time of retirement, they were able to purchase a lot of goods and services from that amount, but due to inflation their purchasing power and standards of living falls.
- Inflation distorts consumer bahaviour. Consumers purchase a lot of goods because of expected future price increases. They hoard goods hoping to 'beat' inflation and in the process create shortages.
- Inflation undermines business confidence. Businesses are unable to make concrete future plans because of uncertainty in price fluctuation in addition, they have to change the price tags on products on a regular basis and this can be so costly and time consuming.
- Inflation and interest rate and savings. The real rate of interest, which is the money rate of interest after making an allowance for inflation, is reduced. Lenders demand for high money rates to compensate for lower real values.

- Lower real interest rates discourage savings and encourage spending. This may have a long-term effect on long-term finance for investment.
- Inflation reduces a country's international competitiveness.
- High prices make products (exports) unattractive on the international market, consumers are likely to prefer cheaper imports to locally produced products. This affects the balance of payments. A country has an adverse balance of payments when exports are lower than imports.
- Inflation causes the currency to depreciate when there is a low demand for exports, therefore, the demand for the currency is low compared to its supply, and the currency depreciates in value.
- Inflation redistributes wealth, it causes borrowers to gain at the expense of lenders as it reduces the value of the debt. The lenders receive less relative to what they had lent. This is related to the time value of money.
- Inflation leads to uncertainty in price forecasting, both at central government level and at corporate business level.
- Money is unable to perform its **functions** properly.
  - Inflation has little impact on money's function as a medium of exchange. Money is still used to purchase goods and services.
  - The use of money as a means of deferred payment is rendered less effective by inflation. Credit is granted but payment is deferred. This leads to redistribution of wealth where borrowers or those who purchase goods on credit gain but lenders lose.
  - The greatest effect of inflation on the functions of money is the function of money as a store of value. The value of money is measured indirectly through prices when prices rise, it is a sign that the value of money has fallen since few items can be brought from the same amount of money. Money becomes an ineffective store of value. Interest rates paid are supposed to compensate, and this is one of the explanations why interest rates rise during periods of inflation.
  - Money is used as a unit of account and as a measure of value. This function is also hampered by inflation as the relative values of things being compared keep on changing in monetary terms.

3.0 UNEMPLOYMENT

Unemployment simply means people do not have jobs. It occurs when people capable of and willing to work are unable to find suitable paid employment.

Unemployment is measured as $\dfrac{\text{\# of unemployed}}{\text{Total workforce}} \times 100$

Full employment is when there are more jobs than people. The number of unfilled vacancies is equal to the number of people out of work. It is the level of national income at which everyone who wants to work is able to do so, in other words, there is sufficient demand to employ everyone.

Classical economists argued that the economy would automatically tend to this equilibrium, due to the market forces of supply and demand. Keynesians maintain that it is the role of government, using policy instruments at their disposal, to ensure that there s full employment in an economy.

3.1 CAUSES OF UNEMPLOYMENT

In some books the words ‘causes’ and ‘types’ are used interchangeably. However there is a distinction.

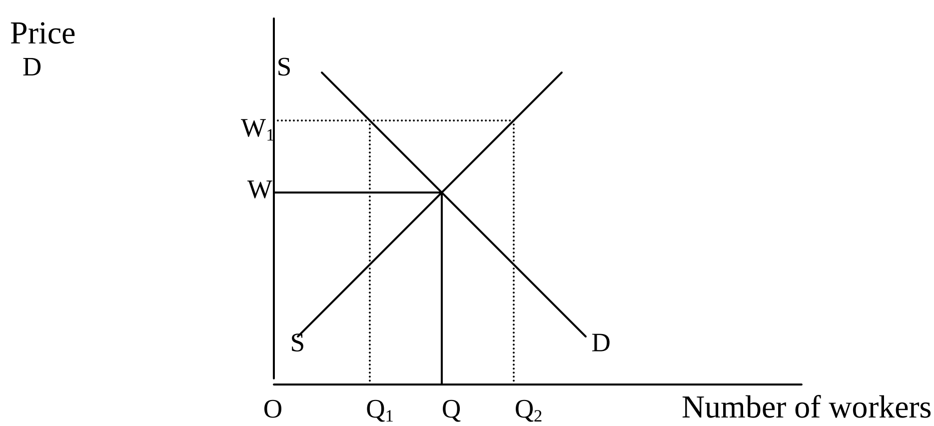
Type is the label given to describe the main common characteristic of some unemployment, while *Cause* is more analytical, an attempt is made to explain how some unemployment has arisen.

Causes of unemployment can be broadly divided into demand and supply factors:

- Demand deficiency unemployment is caused by lack of demand for goods and services, and as a result, firms lay off workers. This is usually when the economy is in the recession stage of the economic or trade cycle and there is little economic activity.

  Keynesians argue that a shortage of aggregate demand is one of the key causes of unemployment.

- Monetarists view Supply side factors such as strong trade unions demanding for high wages as causes of unemployment as firms employ less labour while the supply of labour increases, as shown below:



At a high wage rate of OW_1, the demands for labour by firms reduces to OQ_1, supply naturally increases to OQ_2, because individuals who were unwilling to work at wage rate OW are now encouraged by the high wage rate. The difference between OQ_1 and OQ_2 is unemployment caused by the activities of trade unions.

- Firms lay off workers if *import prices* are too high, like the high price of oil, which reduces a firms' competitiveness, and loss of customers.

- State benefits tend to encourage 'voluntary unemployment'. When the benefits are higher than the market wage, as in the diagram above, a person feels 'better off' being unemployed earning OW_1 than earning a low wage (OW) while in employment.

3.2 TYPES OF UNEMPLOYMENT

- ***Seasonal unemployment*** is considered to be temporal and occurs in certain industries where Economic activity is in specific periods or seasons, examples are tourism, agriculture and construction industries. There is a high demand for labour during certain periods of the year, and then most of the workers are laid off during off peak periods.

- ***Frictional unemploymen****t* is of a short-term duration. It refers to secondary school or college graduates who are searching for jobs, as well as individuals who are in between jobs, the transitional period between workers leaving one job and starting another. Frictional unemployment is also an indication of imperfections in the market such as lack of knowledge, the geographical immobility of labour or a mismatch between the requirements of the employers and the available skills of the unemployed.

  The more efficiently the job market is matching people to jobs, the lower this form of unemployment will be. However, as long as there is **imperfect information** and people don't get to hear of jobs available that may suit them then frictional unemployment is likely to be high.

- ***Structural unemployment*** refers to long-term changes in the pattern of demand and supply in an economy. On the demand side, a firm may fail to compete with rival firms, demand for the company's product declines and the firm is likely to lay off workers and close the business.

  Changes in the supply of a product, for a example if the product like copper ore is getting depleted, there is no need to employ miners and this can also lead to unemployment in the Copperbelt. It may also result from changes in the production methods labour is replaced by machines or capital equipment, termed ***technological unemployment****.* Structural unemployment also includes ***regional unemployment***, some regions in a country may have higher unemployment levels compared to other regions because of different regional economic performances.

  Unemployment results because individuals do not respond quickly to the new job opportunities, they find themselves with no readily marketable talents. Their skills and experiences are unwanted, as they have become obsolete.

- ***Cyclical unemployment*** is the same as ***deficiency in demand unemployment****.* It is characterized by fluctuations in economic growth, characterized by booms and recessions, the trade cycles. During the recession phase, there are high levels of unemployment.

- ***Voluntary unemployment*** is a relatively new concept, defined by the monetarists as being due mostly to high state benefits, either unemployment benefits or being on welfare. This causes people to be unwilling to work at existing low wage rates. They realize that they are "better off" not working and receiving state benefits.

  Voluntary unemployment also includes individuals who simply do not want to work!

3.3 NEGATIVE EFFECTS OF UNEMPLOYMENT

The Economic consequences of unemployment are classified as Economic, financial, social or political costs:

- Labour is a factor of production, and due to unemployment, the Economic resource is not being utilized, this is at a cost, the ***opportunity cost*** of goods and services not produced, quality of workforce diminishes as idleness causes labour to be less efficient, this in turn increases the cost of retraining it.

- Government revenue is mostly from taxes, unemployment results in a loss of government revenue, as the unemployed do not pay any tax, in some rich countries they receive state benefits, which means that unemployment is a financial cost to the government.

- Unemployment may lead to social undesirable behaviour like theft, vandalism, riots or general discontent. The mental and physical health of the unemployed tends to deteriorate, the unemployed are more prone to commit suicide. This is considered to be a social cost.

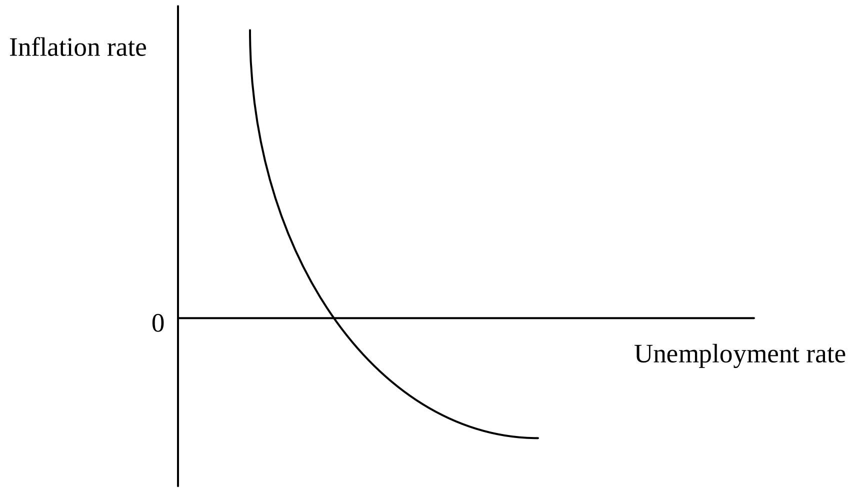
- Whenever there are high levels of unemployment in the country, the political party that forms the government, is likely to lose popularity, this is a political cost to the government.

4.0 THE PHILLIPS CURVE

It shows the relationship between inflation and unemployment. In 1958, Professor A. W. Phillips found a statistical relationship between unemployment and money wage inflation. Inflation and unemployment are two sides of the same coin. If the rate of inflation falls, unemployment rises and vice versa.

Zambia under the United National Independence party (UNIP), was experiencing high levels of inflation, up to three digit figures, but the levels of unemployment were relatively low. Under the Movement for Multiparty Democracy (MMD), the country has experienced low levels of inflation but very high levels of unemployment.

The Phillips Curve explains the "trade off" between inflation and unemployment; it is a graphical illustration of the inverse relationship between inflation and unemployment. It shows that the lower the rate of inflation the higher the rate of unemployment.



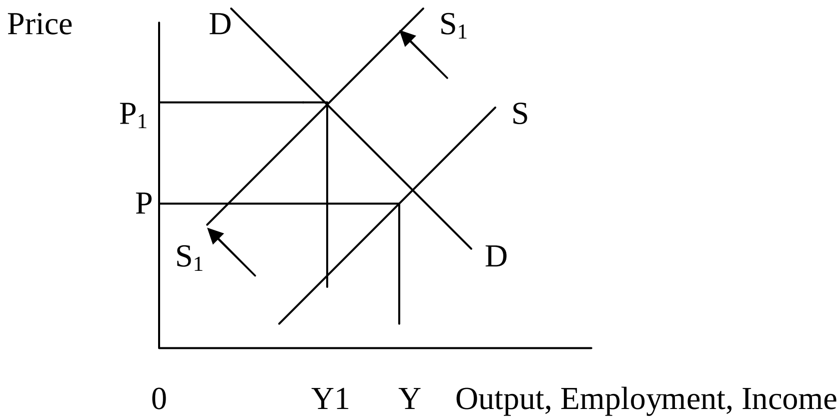
High inflation is associated with low unemployment. Note that the curve crosses the horizontal axis at a positive value for the unemployment. It is not possible to have both zero inflation and zero unemployment, zero inflation is associated with some unemployment.

The above means that the government cannot achieve two of its macroEconomic objectives of low rates of inflation or stableness and low rates of unemployment at the same time. The two are mutually exclusive. The government can only achieve one objective at the expense of the other.

4.1 STAGFLATION

Once in a while, in any country the “trade off” does not apply, as both inflation and unemployment move in the same direction. This situation is known as stagflation.

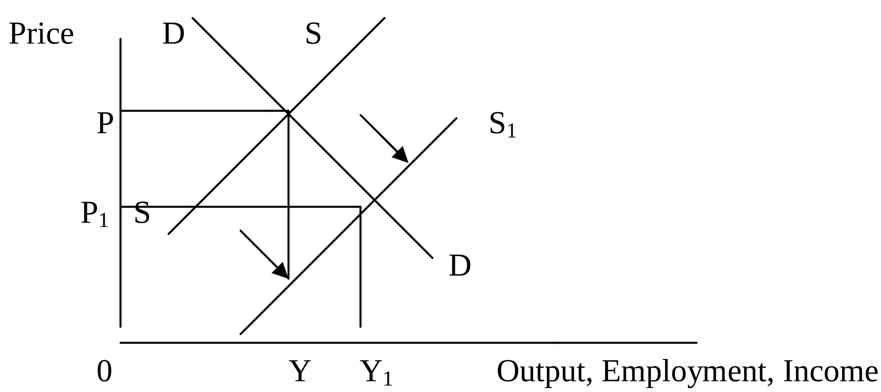
Stagflation is a term coined by economists in the 1970s to describe the unprecedented combination of slow Economic growth and rising prices. The Phillips curve does not apply, there is no “trade off”, and instead, there are unacceptably high levels of both inflation and unemployment. This means a country can be experiencing stagnation, the recession phase of the trade cycle and very high levels of price increases.
The a above maybe as a result of high costs of production, especially the price of crude oil, which may cause the supply curve to shift to the left. This causes the price to increase from 0P to 0P1 and output, employment and income reduces from 0Q to 0Q1



5.0 SUPPLY-SIDE POLICIES

Monetarists believe that stagflation is as a result of ignoring the aggregate supply side of the equation on supply and demand analysis. Keynesians believe in manipulating aggregate demand in order to manage the national economy, and monetarists argue that Keynesian demand management is inflationary, the solution is to put in place measures to improve the supply of goods and services, known as supply side policies.

Supply-side policies can be used to reduce market imperfections. This should have the effect of increasing the capacity of the economy to produce, that is increase output, employment and income and reducing prices at the same time. It is without doubt the only non-inflationary way to get increases in output.

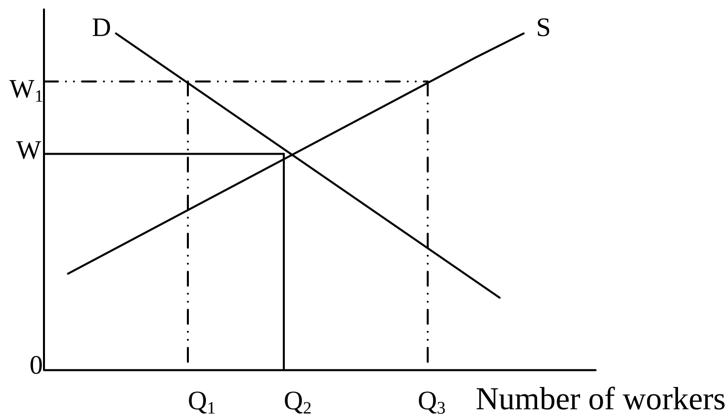


The idea is to increase aggregate supply from SS to S_1S_1 in order to increase output to Y_1 and at the same time reduce prices from 0P to $0P_1$.

The above is an indication of the need to shift both Economic and government policy towards supply side policies. The long run Economic growth and standard of living are both functions of both production and supply. The low prices from increased supply imply that a country can compete with the low cost producing countries of South East Asia.
In general, supply side policies aim to remove market imperfections and encourage individualism in order to increase efficiency and raise competitiveness. They are micro orientation, unlike Keynesian policies that are macro.

Some of the best-known supply side policies are:

- Lower income taxes. High direct taxes are a disincentive to enterprise and hard work, more especially overtime. There is need to encourage individuals and firms to be more enterprising, and to increase production.
- Privatization and deregulation, since government intervention and regulation weakens a country's ability to make the economy dynamic and self regulating, Adam Smith's 'invisible hand' in the market. Public provision of services, government grants and subsidies encourage inefficiencies, and state owned industries are not competitive.
- Strong trade unions and employment legislation lead to unemployment and encourage over manning. There is need to have weak trade unions and workers who will accept 'flexible' wages.



Strong trade unions can successfully bargain for high wage rates ($0W_1$), which results in few workers ($0Q_1$) being employed, while the supply is high at $0Q_3$. By accepting lower wages (0W), more workers would be in employment ($0Q_2$). The inflexibility in the labour market creates unemployment.

- Related to the above, are wage controls, wage regulations and employment legislation which all contribute to inflexibility, workers 'pricing themselves' out of the market and ultimately unemployment. According to the supply side policies, these should be abolished.
- Better information on job opportunities and adequate training is what is required for the aggregate supply curve to shift to the right.

6.0 CHAPTER SUMMARY

Inflation to a layman is simply a sustained increase in the price of goods and services. Inflation is measured as a percentage change, and the two extremes are creeping inflation to hyperinflation. Inflation can be caused by demand factors, supply factors, or according to the monetarists, any change in the money supply is inflationary.

There are several reasons why inflation is considered to be economically undesirable, it affects planning both at central government and at corporate business level and it also undermines business confidence. Inflation reduces a country's international competitiveness and causes the currency to depreciate given a low demand for exports. Inflation discourages savings, and ultimately, investment. It also distorts consumer behaviour, consumers purchase a lot of goods in the hope of 'beating' inflation. More importantly, inflation has a big impact on people who are on fixed incomes, their purchasing power and standard of living **falls**, and money is unable to perform its **functions** properly.

Unemployment simply means people do not have jobs. The words 'types' and 'causes' of unemployment are usually interchanged, but generally, unemployment is categorized as cyclical, structural, seasonal, frictional and voluntary. Unemployment also has a number of negative consequences, classified as Economic, financial, social or political.

A government can control both inflation and unemployment using either fiscal or monetary policies, or both. Unfortunately, there is a negative relationship between inflation and unemployment, which is illustrated by the Phillip's curve. The government has to 'trade off' inflation for unemployment or vice versa. Sometimes, there is an increase in inflation and unemployment, a situation known as stagflation.

An effort to 'cure' both inflation and unemployment is explained by the monetarists using the supply-side policies. These policy measures are intended to free up the supply of goods and services in all markets, eliminating market distortions, increasing production, the 'supply' side of the equation, through deregulation. The government has to reduce taxes, privatize, allow the labour supply to move freely, weaken trade unions, etc.

CHAPTER 6

PUBLIC SECTOR ECONOMICS

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After studying this chapter, the students should be able to:

- Appreciate the role of government in the economy.
- Identify the main items of government expenditure
- Explain of the different sources of government revenue and
- Understand the advantages and disadvantages of the main sources of government revenue
- Understand fiscal policy
- Identify the fiscal stance through the public sector net cash requirements
- Explain of the nature and composition of national debt

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1.0 Introduction

There are many aims to public finance, but the priority on how the government deals with its finances in the course of performing its functions varies with political complexities. However, generally, the bulk of government revenue is spent on socially desirable expenditures such as education, health and social services. Therefore, public finance is concerned with government expenditure and government revenue, and the difference between them, which is the public sector net cash requirements.

2.0 Government expenditure

Government capital expenditure refers to government spending on investment goods. This means spending on things that last for a period of time. This may include investment in hospitals, schools, equipment and roads.

Government current expenditure refers to government day to day spending. This means spending on recurring items. This includes salaries and wages that keep recurring, spending on consumables and everyday items that get used up as the good or service is provided

In general, the main items of government expenditure in most countries can be can be classified under the following headings;

- Defence
- Internal security that is, the police, fire brigade etc.
- The merit goods under the social sector like education, health and housing.

- Economic policy covering subsidies to agriculture and industry, the provision of capital to the nationalised industries (government investments).
- Social security and other transfers make up a big chunk of government expenditure in developed countries
- Debt interest payments on the national debt are a big burden for the poor countries.
- Miscellaneous expenditure such as diplomatic services.

2.1 Government revenue

This is mostly from taxation. However, taxation has other functions besides covering central and local government expenditure.

The other reasons for taxation are:

- To check the consumption of demerit goods like beer and cigarettes and to cause the pricing of the products to reflect the social costs to society of smoke related illnesses.
- To reduce inequality of incomes and wealth through a progressive system of taxation.
- To put into effect the 'automatic stabilisers', that is increase the levels of direct taxes to dampen the upswings and reduce on the inflationary pressures when the economy is at the 'boom' phase of the trade cycle.
- To protect infant, strategic and declining industries by introducing indirect taxes like import duties to discourage imports by making them more expensive and therefore less competitive.

2.2 Principles of taxation

Adam Smith outlined the basic characteristics of a good tax system as the four canons of taxation, namely:

- Equity, which means that taxes should be fair and therefore should depend on
- an individual's ability to pay. Taxes must be proportional to one's income.
- Certainty, with regard to the amount to be paid, how, where and when it should be paid.
- Convenience of payment and collection by the taxpayer.
- Economy, that is, the cost of collection should not be excessive especially in relation to yield.

The additional principles of taxes considered by governments are summarised as **efficiency** and **flexibility**. Taxes must as far as possible achieve its objective efficiently and not undermine other aims and taxes. It should also be adjustable to changes in policy.

2.3 Classification of taxes

Taxes can be classified in several ways depending on:

- Who is levying the tax? This can either be the Central or the Local government.
- What proportion of a person's income is taxed? There are three categories:

a) A progressive tax

A progressive tax is a tax that takes an increasing proportion of income as income rises. The rate of tax keeps on increasing with every subsequent increase in income. Most direct taxes are progressive, a good example is income tax, and the rate increases as a person earns more.

b) A regressive tax

This takes a higher proportion of a poorer person's income. Most indirect taxes are regressive. A regressive tax is a tax that takes a smaller proportion of income as income rises, this means it takes a higher proportion of a poorer person's income. In other words it is a tax that hits less well-off people harder than the better off. Most indirect taxes are regressive. An example of a regressive tax is the television licence. It is exactly the same amount for everyone, which makes it a much smaller proportion of a large income than a small one.

c) A proportional tax

This is when the tax is the same proportion on all incomes, whether large or small. It simply taxes a given proportion of one's income for example 10% of K500,000.00, 10% of K5,000,000.00 and 10% of K50,000,000.00.

Tax burdens that are proportion to income are considered to be fair, however, they do not contribute towards equal distribution of wealth.

- Who is paying the tax? Is it a direct or an indirect tax?

  A **direct tax** is a tax on income, profit or wealth. It is paid directly to the revenue authorities by the taxpayer. Examples of direct taxes are

  - Income Tax
  - Corporation Tax
  - Capital Tax
  - Inheritance tax
  - Other taxes to the local government like personal levy, motor vehicle duties

  Advantages of direct taxes:

  1. Are equitable, that is they conform to the principle of 'ability to pay', through the progressive system of taxation.
  2. Have an elastic and high yield; the rate of taxation can be increased and therefore increasing government revenue.

3. Are certain, both the taxpayer and the government know the amount to be paid, how, where and when it should be paid.
4. Lead to equal distribution of income and wealth, this again is through the progressive system of taxation.
5. Are automatic stabilizers through their progressive nature, taking more money out of the economy (withdrawals) when the economy is in its 'boom' phase, while taking less money and increasing welfare payments when the economy is faced with a depression.
6. Are not inflationary like indirect taxes.

Disadvantages of direct taxes

1. High rates acts as a disincentive to efficiency, effort and enterprise. Tax reduces the return on the investment and reduces a firm's ability to invest and expand as this depends on the retained profits. Workers are less inclined to put in extra hours. Individuals and firms want to make money for themselves and not to contribute to government revenue.
2. High rate might also encourage migration of skilled manpower to 'tax havens'
3. High rates encourage tax avoidance. People find loopholes so as to avoid paying tax. It also encourages tax evasion, which is illegal non-payment of tax especially in the informal sector.

An **indirect tax** is a tax on expenditure. Tax is paid indirectly to the revenue authorities as part of the payment for a commodity or service, whenever particular purchases are made. Examples of Indirect taxes are:

- Customs or import duties
- Excise duties, this is a tax on some locally produced commodities
- Value added tax.

The **advantages** of indirect taxes:

1. Revenue yield from indirect taxes help to avoid high direct taxes.
2. Payment is certain since they are difficult to avoid and to evade.
3. Convenient to the taxpayer since they are paid in small amounts and at intervals instead of one big lump sum of money which is deducted and paid every month like pay as you earn. In addition, they are convenient in that they are paid when an individual is in a position to buy the commodity and therefore can afford to pay the tax.
4. Economical in collection as companies and traders collect on behalf of the government and reduce the administrative burden that should fall on the revenue authorities.
5. It is not harmful to effort and initiative like direct taxes. Instead, it is less painful since it is hidden in the price of a commodity or service.
6. Most importantly, indirect taxes are flexible instruments of policy as they can be adjusted to specific objectives of Economic policy such as:-

    - Protecting infant industry or vital (strategic) defence industries
    - Strengthening political links e.g. Southern African Development Cooperation (SADC) and Common Market for East and Southern Africa (COMESA).

- Citizen’s health may be safeguarded as indirect taxes can be used to encourage or discourage the production and consumption of particular goods and services.
- The balance of payments may be strengthened or improved by taxing certain imports.

The **disadvantages** of indirect taxes:

1. There are regressive, **a flat rate** like a poll tax, a **specific tax** charged as a fixed sum per unit sold or an **ad valorem tax** which is charged as a fixed percentage of the price of the good with no concessions for people in the low income bracket, take a higher proportion of the income of low income earners than high income earners.
2. They do not depend on a person’s ability to pay both the rich and the poor pay the same amount as tax as long as they both buy the same product or service. Therefore they are not equitable.
3. Some people who use unauthorized border entry points, the ‘black economy’, may evade indirect taxes like customs duties.
4. May encourage inflation, whenever value added tax, customs duties or excise duties increase, the prices of taxed goods and services also increase.
5. Possibly harmful to industry especially for goods with elastic demand

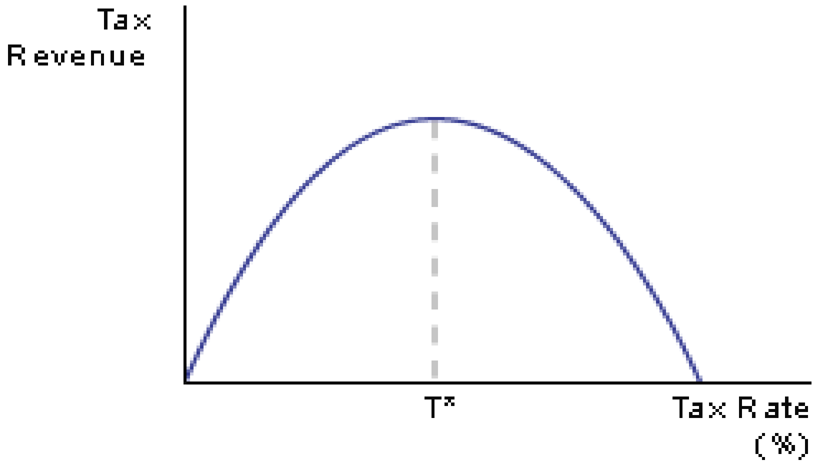
3.0 LAFFER CURVE

Government revenue is mostly from taxes. A Laffer curve shows how tax revenue and tax rate are related.

The Laffer curve is named after Professor Art Laffer who suggested that if the tax rate is 0%, then government revenue would be zero. If the tax rate is 100%, again there would not be any government revenue as individuals and firms would not be willing to contribute 100% of their income to the government. No one would be willing to work.

The Laffer curve also shows the rate at which the government can achieve a maximum revenue Tr, the tax rate should be Tx. In addition, it also shows that the government can achieve very high tax revenue at two rates, 25% and 75%.

Given the fact that a high tax rate discourages hard work and enterprise, the best option is a tax rate of 25%. According to the advocates of supply side policies, lowering tax rates increases production and supply. This in turn increases the national income.



T^x represents the optimum tax rate where the maximum amount of tax revenue can be collected.

4.0 PRIVATISATION VS NATIONALISATION

Privatisation implies
- The transfer of the nationalized industries to private ownership.
- Selling state assets, either completely or partially
- Opening up state monopolies to outside competition
- 'Contracting out' to the private sector services paid for out of public funds, such as, refuse collection, which was previously done by the local government.
- Charging beneficiaries 'Economic fees' for publicly provided goods and services like hospitals and schools.

Therefore, privatization implies more than the movement of assets from the public to the private sector. It embraces all the different means by which the disciplines of the free market in the provision of goods and services can be applied to the public sector.

The case for privatisation is the argument that is put forward for **deregulation** of industries, which is, the removal or weakening of any form of state interference with the operation of free market activity

The main aim of deregulation/privatisation is to improve competition and efficiency.
Once the statutory barriers are removed, the economy is said to have **liberalized** industries or it is following a **liberalized** market economic system, compared to the command economic system.

4.1 Arguments for privatisation

a) Reduced burden on the public purse as the government no longer supports loss- making nationalized companies. Privatisation allows a reduction in the public sector borrowing requirement and tax cutting, as it provides funds for the treasury when companies are sold.

b) There is greater economic freedom from detailed economic control as privatized companies are not subject to state control.

c) Improved efficiency through competition in the market, this encourages producers to cut their costs in order to be more competitive, and firms have to be innovative in the search for profits.

d) In addition to the above, there is also improved quality since firms have to compete to survive and have to be responsive to customer complaints.

e) If companies are not in state control, there is greater resistance to the power of trade unions, industries are more fragmented and difficult to organise.

f) Privatisation leads to a creation of a property-owning class, more people are able to buy shares, this gives buyers market power, they work harder and strike less, a better understanding of private profit motive and business problems.

g) Costs and inefficiency decrease as bureaucracy from nationalized companies is reduced.

4.2 Arguments against privatisation

a) Privatisation does not mean that competition is automatically enhanced. Instead, private monopolies have been created. An example is if the Zambia Electricity Corporation (ZESCO), became privatized, it means a previously government controlled monopoly becomes an uncontrolled one in private hands, with no public responsibility. Consumers may suffer. However, this is what leads to most governments to regulate or attempt to regulate the newly privatized companies in much the same way as the nationalized industries.

b) Just as privatization does not mean competition, it also does not guarantee efficiency. Customers have ended up with fewer services, and at higher prices. A good example is rural transport. The government owned United Bus Company of Zambia (UBZ), used to go to all the rural areas, everything was timetabled (date and time).

c) The quality of service has reduced, with costs being saved by reducing the number of workers ‘right sizing’, paying lower wages and reducing the services that were being provided, as mentioned above, some routes were termed ‘unprofitable’ or the roads ‘impassable’.

d) Privatisation may allow people in rural areas without Economic power to suffer, since loss-making services are not provided by the private sector, most of which are important to the poorest members of the society.

e) In theory, it is the loss-making companies that are supposed to be privatized, but in practice, the privatization exercise is rarely properly done in most countries in the world, for example asset sales are under priced to attract buyers and in the process, create big capital gains for private investors.

f) Companies that are in private hands often pay their top executives very large salaries and offer them very good conditions of service, while reducing the powers of trade unions and paying union members low wages. This lowers the morale of the workers and lowers productivity while encouraging pilfering, strikes etc.

g) If competition is enhanced through privatisation, it sometimes leads to waste of resources and the duplication of goods and services, an example is monopolistically competitive market structures.

4.3 Nationalised industries

The public sector includes some businesses run by the government, such as Zambia Electricity Company (ZESCO), Zambia Telecommunications Company (ZAMTEL), Lusaka Water and Sewerage Company etc. Managers of such companies are accountable to the elected politicians (ministers) in charge of that sector, and government-sponsored boards such as the Zambia Energy Regulation Board, which regulates ZESCO, regulate them.

The **case for nationalization** can be considered alternatively as the disadvantages of liberalisation.

a) Nationalisation can lead to reduced costs through economies of scale, since with increased competition, each firm produces less output on a small scale, and unit costs increase.
b) There is provision of un economic services for consumers. Nationalisation, just like the socialism or planned economic system, social benefits are placed above private profits. It considers the net gain to society, to the point of keeping industries that are clearly technologically inefficient, as in the case of Maamba coalmines. Another argument in favour of providing uneconomic services is that it helps to protect employment.
c) It is sometimes in the national interest that some basic industries are brought under public control, especially, strategic industries which would be dangerous under private ownership such as atomic or nuclear energy.
d) It may also be necessary to carry out government policy, like controlling the money supply, as in the case of the Bank of Zambia.
e) Nationalised industries have sufficient capital available for investment, because of government support. Where competition is wasteful, it maybe better to create a large state-owned monopoly, to avoid waste and duplication.
f) A fairer distribution of wealth, the huge profits do not go to the capitalist owners, surpluses are used for the benefit of society. A case in point is ZESCO, the supernormal profits are used for rural electrification. The supernormal profit also justifies the high salaries enjoyed by ZESCO employees.

5.0 FISCAL POLICY

Fiscal policy is the use of government expenditure and taxation to try to influence the level of economic activity. It is the decisions and actions of the government regarding its expenditure and its revenue taxes, that is, since government revenue is mostly from taxation. The government as an instrument of economic policy uses fiscal policy, also known as budgetary policy, through the balance between government expenditure and revenue. In order to reduce **high** levels of unemployment, or to stimulate recovery from a recession, the government aims for a **budget deficit** or an **expansionary** fiscal policy.

An expansionary (or **reflationary**) fiscal policy could mean:

- Cutting levels of direct or indirect tax
- Increasing government expenditure

Reducing taxes causes government revenue to be lower than government expenditure, which results in a budget deficit, government borrowing covers the difference. The government needs to borrow money to finance its activities. This borrowing is referred to as the **public sector net cash requirement** (PSNCR). It used to be called the **public sector borrowing requirements** (PSBR). The PSBR is the amount of money the government needs to borrow to meet their spending plans. In other words it is the amount that their spending exceeds their **tax revenue.** Therefore, an increase in the PSNCR or PSBR is a sign that the government is following an expansionary fiscal policy. The effect of expansionary policies would be to encourage more spending and boost the economy.

Budgetary policy can also be used to check inflation or an adverse balance of payment by aiming for a **budget surplus**, or a **contractionary** fiscal policy. This is the exact opposite of an expansionary policy.

A contractionary (or **deflationary**) fiscal policy could mean:

- Increasing taxation, either direct or indirect
- Cutting government expenditure.

Reducing government expenditure while increasing taxes is what leads to a government surplus. The difference is the public sector net cash surplus. This used to be called the **public sector debt repayment** (PSDR) the government is in a position to service the debts plus interest. It is a sign that the PSNCR are low. A reduction in both government and consumption expenditure reduces the level of demand in the economy and help to reduce inflation.

5.1 The Public Sector Net Cash Requirements (PSNCR)

The balance between government expenditure and government revenue shows the fiscal stance being followed by the government. Government expenditure is an injection into the circular flow of income, a component part of aggregate demand, therefore an increase in government expenditure is an indication of the expansionary stance being followed by the government. Whereas taxation is a withdrawal or leakage from the circular flow of income, and as such, increasing taxes is an indication of a contractionary stance.

In practice it maybe difficult to reduce the growth rate of public expenditure due to for example political factors such as by-elections and defence if a country is at war. Existing capital projects, which can only be, completed over a period of years, as well an economic depression, which results in high welfare and unemployment benefits to act as automatic stabilisers. If the size of the PSNCR increases from one year to the next, then it is a sign that the government wants to boost the economy.

In general, for most rich nations, the increase in the PSNCR can stem from the increased life expectancy and an ageing population, which implies more spending on social security. High unemployment levels which may lead to more unemployment benefits being paid. For most poor countries, an increase in the PSNCR can stem from debt interest, political commitments like the numerous by-elections in Zambia or high inflation levels which raises the cost of public provision of goods and services.

Both the developed and the developing countries can be affected by tax rate changes, tax reductions for example, reduce government revenue, so the government has to depend on borrowing.

5.2 Parliament control procedures

The legal framework that governs the management and control of the public finances in Zambia is made up in the constitution. Parliament is supposed to provide the necessary checks and balances in the budget process. During the fiscal year the scrutiny of the spending reports, if any, is done by the Committee of Supplies, while hearings on expenditures is conducted by the sessional committees of Parliament. The Minister of Finance and Economic Planning must prepare supplementary estimates for expenditure, for approval by the National Assembly within a period of four months and if the National Assembly is in recess at the first sitting of the Assembly. The Minister of Finance is required to prepare a Supplementary Appropriation Bill confirming the approval by Parliament of such expenditure or the excess of expenditure within 15 months after the end of the financial year. If the National Assembly is not sitting then, the bill must be tabled within a month of the first sitting, *a Bill to be known as the Excess Expenditure Appropriation Bill*.

Thus, the roles and responsibilities of the legislature is moderately well assigned in principle but the budget disbursement of resources to spending units is appropriated on the basis of ad-hoc criteria which can later be legitimised by both supplementary and excess expenditure acts, and there is inadequate time for parliamentarians to scrutinise budget documents.

5.3 GOVERNMENT REFORM PROCESS (from Public Expenditure Management and Financial Accountability, PEMFA Evaluation Report)

5.3.1 General description of recent and on-going reforms

In the early 1990s, Government began a political and socio-Economic reform process, which entailed democratising the political system and liberalizing the economy. The political reforms gave special impetus to public demand for good governance, transparency and accountability in the conduct and management of public affairs. The Economic reforms focused on privatization of parastatal entities and the redefinition of the role of the Public Service from that of controlling the overall economy to that of providing a conducive environment for market based and private sector driven economy.

5.3.2 Public Service Reform Programme (PSRP)

In 1993, Government initiated the PSRP to restructure the Public Service in order to improve the quality of service delivery. Therefore, Government designed the Public Service Capacity Building Project (PSCAP) as a comprehensive strategy to build institutional and human capacity for quality public service delivery, it became operational in October 2000 and was designed to be implemented over a thirteen-year period (2000-2013). The focus of the first phase was on the following five major outputs:

- right-sizing and pay reform of the Public Service,
- improved policy and Public Service management,
- improved financial management, accountability and transparency,
- improved capacity of the judicial and legal systems and
- decentralisation and participatory governance.

5.3.3 Poverty Reduction Strategy Paper (PRSP)

The PRSP was developed as the Nations' medium term overall policy framework for national planning and interventions for development and poverty reduction for the period 2002-2004. The strategy for poverty reduction was rapid economic growth and employment creation. This would result in improvements in national resources management, a conducive macroeconomic framework, sectoral performance improvements especially in key sectors such as agriculture and social sectors, infrastructure developments, overall improvements in governance and public service delivery capacity.

6.0 AUTOMATIC STABILISERS

During a recession 'phase', income does not fall to zero because the benefit (welfare and unemployment benefits) system provides some income. The effect of automatic stabilizers when an economy is recovering from a recession is known as 'fiscal drag'.

Fiscal drag refers to the effect inflation has on average tax rates. If tax allowances are not increased in line with inflation, and people's incomes increase with inflation then they will be moved up into higher tax bands and so their tax bill will go up. However, they are actually worse off because inflation has cancelled out their pay rise and their tax bill is higher, this is to the benefit of the revenue authorities. It is getting more tax without increasing tax rates, a subtle means of raising more tax revenue. To maintain average tax rates, allowances should be increased by the amount of inflation each year.

7.0 THE PUBLIC DEBT

This is the total amount of accumulated borrowing by the local and central governments including public corporations, to its various creditors both local and foreign, the International

Monetary Fund (IMF), the World Bank etc. Note that debt increases, interest rate payments form a large portion of government expenditure.

The debt instruments are of two types:

Debt management in terms of contracting, servicing and repayment is a major element of the overall fiscal policy. The financial report of the Minister of Finance and Economic Development must include a statement showing the particulars of debt charges paid in that financial year in respect of loans raised under the Act. A loan may be raised as a debt instrument of two types, **marketable debt,** this is either short-term debt that consists of treasury bills or long-term debt that consists of Government bonds or by agreement in writing. **Non-marketable debt** consists of any other debt raised by the Government either internally or externally. In addition, the debt can be **reproductive**, that is, used to purchase a real asset or it can be a **deadweight** debt, meaning that no assets are covering the debt.

The Act empowers the minister to raise any loan in accordance with such conditions and upon such terms, as s/he shall direct. If the loan is raised through the issue of a bond, stock or Treasury bill, the Bank of Zambia is the Minister's agent.

The Zambian National Debt problem is being addressed with the implementation of the HIPC Initiative (from multilateral and Paris Club bilateral creditors), voluntary additional relief on part of some of the Paris Club bilateral creditors and the G8 debt cancellation initiative.

Prudent fiscal policy/discipline needed to reduce the budget deficit to manageable levels.

8.0.0 FIFTH NATIONAL DEVELOPMENT PLAN (FNDP)

Governments accept the Keynesian Theory that active Government involvement in the economy is necessary for **macroEconomic stabilisation**. In a mixed Economic system, the party in power can change the shape of the economy. The Government, it may decide to trim down the public sector and fatten the private sector, or vice versa.

The Zambian government has a major Economic role and responsibility, it has articulated its long -term development objectives in the National Vision 2030, and the FNDP is an important step towards the realisation of this vision. The development goals are:

(a) Reaching middle-income status
(b) Significantly reducing hunger and poverty
(c) Fostering a competitive and outward oriented economy.

The theme of the FNDP is 'broad based wealth and job creation through citizenry participation and technological advancement'. The broad MacroEconomic objectives for the FNDP are as follows:

- To accelerate pro-poor Economic growth, this is the main goal of the FNDP, to realise this goal, the aim is to have an annual growth rate of at least seven percent, and ensure that growth is broad based and rapid in the sectors where the poor are mostly engaged.

- To achieve and sustain single digit inflation
- To achieve financial and exchange rate stability
- To sustain a viable current account position, and
- To reduce the domestic debt to sustainable levels.

The above are the overall characteristics of the economy, which any government discern as desirable. An additional objective, which is included in the FNDP theme of job creation, is attainment of full employment.

The MacroEconomic objectives of Governments are interdependent, at the same time, simultaneous success is impossible (Phillips curve!). The FNDP contains the policy instruments that the Government intends to use to achieve the MacroEconomic objectives listed above, are outlined as follows:

8.0.1 FISCAL POLICIES

To focus on avoiding excessive fiscal deficits and debt by reducing government borrowing which in turn, contributes to a decline in interest rates, besides the reduced external debt servicing through the highly indebted poor countries (HIPC) initiative. This will also allow for an expansion of credit to the private sector, no **crowding out** effect!

However, budget execution need to be improved, financial accountability and expenditure monitoring systems need strengthening, as well as strengthening the revenue base, whose weak systems have created potential avenues for fraud.

8.0.2 MONETARY AND FINANCIAL POLICIES

To focus on achieving and maintaining single-digit inflation as a pre-requisite for reducing high interest rates which have contributed to poor access to financial services within the economy. The Zambian financial sector is characterised by high cost of borrowing, thin capital markets and absence of financial services in rural and peri-urban areas. The focus in the FNDP is to develop the capital markets, developing and implementing a rural financing policy and strategy and the strengthening of banking and non-banking financial institutions.

8.0.3 SUPPLY-SIDE POLICIES

To reduce inflation, there is need to address supply side factors such as the poor infrastructure and marketing systems, the vulnerability of the agricultural sector to weather fluctuations, and weak policy implementation.

8.0.4 EXTERNAL SECTOR POLICIES

The objectives of the external sector are to achieve the following:

- Sustain a viable Current Account balance by promoting export growth and maintaining a competitive exchange rate. The mining sector will continue to play an important role, however, the development strategy focus is diversification away from copper.
- Improve the external competitiveness of the economy.
- Maintain a sustainable external debt position.

8.1 POLICY CONSTRAINTS

In practice, Governments have limitations in their ability to achieve MacroEconomic objectives generally, because of various reasons. Some of the constraints are:

a) **Previous policy decisions** taken by the previous government, especially the fiscal and regional policies.
b) The Government may **lack perfect knowledge/**information of the economy. The statistics may be outdated or based on estimates.
c) Policy changes take time to implement and time to be effective, **a time lag**.
d) There is **no ceteris paribus**, this means that there is no technique that to hold other variables constant. Zambia has extremely free trading with borders very wide open compared to countries like Zimbabwe and South Africa.
e) **Political factors** often supersede prudent policy Economic judgement, especially in a developing country like Zambia with not enough checks and balances and unplanned by-elections!
f) In addition to the constraints of information and time, the **methods may be inefficient** in that they do not achieve their targets, or the pursuit of one policy instrument may limit the effectiveness of the other.
g) The fluctuating patterns of booms and recessions**, the trade cycle**, can affect the achievement of macroEconomic objectives. For example, when there is international recession, there is no Economic growth.

9.0 CHAPTER SUMMARY

Public finance deals with finances of the Government, which is reflected in terms of expenditure and revenue. Governments spend their income on the provision of a variety of services that the private sector does not provide. Examples are defence, internal security, education, health etc.

The large sums of money, which governments require, are obtained primarily through taxation levied on incomes (direct, progressive taxes), and on goods and services sold (indirect, regressive taxes). Taxes may be apportioned among people on the basis of ability to pay or on the basis of benefits received. Each type of tax has its advantages and drawbacks.

Another source of government revenue is through privatization. This is the transfer of assets from public to private ownership. However, it has some advantages and disadvantages, hence there are some arguments in favour of nationalized industries.

When it is inexpedient or impossible to raise needed funds, governments may borrow money, either on a long or a short-term basis. In addition, governments manipulate their tax impositions, their expenditures and their borrowing so as not to merely to finance desirable projects and services but also to maintain the national economy in a stable condition, known as fiscal policy. The objective of fiscal policy can be either a deflationary gap, operate a budget deficit (reduce taxes and increase government expenditure) in order to reduce the high levels of unemployment. Alternatively, a government can aim for an inflationary gap, that is, operating a budget surplus (increase taxes and reduce government expenditure) in order to check inflation.

The total amount of government borrowing is known as the national debt.

The FNDP is an important step towards the realisation of national vision 2030, and the development goals are:

a) reaching middle-income status
b) significantly reducing hunger and poverty
c) fostering a competitive and outward oriented economy.

The government came up with five discernable macroeconomic objectives, these will achieved using various instruments such as fiscal and monetary policies. In practice, governments face a number of policy constraints which hinder them from achieving the macroeconomic objectives.

CHAPTER 7

INTERNATIONAL TRADE

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After studying this chapter, the students should be able to:

- Appreciate the growing impact of globalisation and the multinational companies
- Explain why countries undertake international trade and the benefits they get from international trade
- Understand the law of absolute and the law of comparative advantages
- Distinguish free trade from protectionism
- Identify the reasons for trade restrictions, and the different forms of trade restrictions
- Understand terms of trade and its importance
- Appreciate of international organizations that facilitate free trade

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1.0 INTRODUCTION

International trade involves the exchange of goods and services between countries. It involves trades among nations. A nation trades because it lacks the raw materials, climate, specialist labour, capital, or technology needed to manufacture a particular good. Thus, international trade arises because countries have different production capacities and different demands for goods and services.

1.1 GLOBALISATION

Economic activity has been internationalized. This is reflected in the growth of trade and other capital flows, currency bought and sold in the foreign exchange market, has lead to the term **global economy**. The global economy refers to an open economy where the ratio of exports to output forms a significant proportion of economic activity.

World trade has been expanding to the extent that neighbouring countries that have always traded with each other are making such arrangements more formal. Trade agreements like the **free trade areas** where countries agree to reduce or abolish trade restrictions between member countries, while allowing members to impose their own separate trade restrictions against non-member states.

Alternatively, it may be extended into a **customs union**, where free trade is encouraged among members but erect a common external tariff on imports from non-member states. In addition, there are **common markets**, which are similar to customs unions but include the free movement of **factors of production** as well as trade.

1.2 MULTINATIONAL COMPANIES (MNCs)

The growth of multinational companies has taken place in an environment of increasing globalisation of markets.

A multinational company is one, which owns or controls production or service facilities in more than one country. Note that a company does not become a multinational simply by trading internationally!

The growth of globalisation is unstoppable, and with it is their power to influence international trade. However, the extent to which the Zambian economy benefits from MNCs is difficult to assess.

The assessment of the impact of MNCs on national economies is considered under various costs and benefits. Direct foreign investment by an MNC should improve economic welfare as capital is transferred, capital inflow into a relatively poor country, and it should also promote technology transfer. New technologies being transferred without the research and development costs.

In addition, local companies can copy superior processes and organizational patterns. Employment can also be provided.

In practice, MNCs only transfer technologies at low levels, and once the profits from the investment are remitted back, it becomes an outflow of foreign currency.

Most MNCs may decide to employ their own nationals in top management positions. The worst part is that MNCs are offered grants, subsidies, tax relief etc., in order to attract them into poor economies like the Zambian one, while MNCs can gain a cost advantage, integrating vertically by establishing assembly plants in countries where there is abundant cheap, high-quality labour.

Some MNCs pursue a policy of horizontal integration in order to gain new markets and expand sales. The advances in communication, cheap air travel, development of satellite systems has made it cheaper for MNCs to develop new markets in overseas countries.

2.0 REASONS FOR INTERNATIONAL TRADE

- Products that are not produced in a certain country are available in other countries, thanks to international trade. For instance, computers are not produced in Zambia, but are produced in the United States.

- Unequal distribution of skills and technology. In addition, some countries have a good reputation in the production of some commodities than other countries. An example is a country like Japan which is more skilled in the production of goods like cars.

- Excess demand for locally produced goods may force countries to import to offset the shortage.

- Unequal distribution of resources. For example, oil is found in Angola but not in Zambia, the climate in South Africa is suitable for growing apples, but not the Zambian climate, etc.

2.1 THEORY OF COMPARATIVE ADVANTAGE

Comparative advantage is a country's ability to produce a product at a lower opportunity cost in terms of another country. The principle of comparative advantage states that countries will benefit by concentrating on the production of those goods in which they have a relative advantage.

A country is said to enjoy a comparative advantage over another if, with the same input of resources, it can produce more of a good than another. A nation's comparative advantage is measured in relation to all goods and services it produces. A country has a comparative advantage in those products that it can produce cheaply.

Any product can be produced in any country but what matters most is the cost of production. It is therefore more beneficial for each country to use its resource in the production of those goods in which it has a cost advantage, and to trade with other countries to obtain, those goods, which cannot be produced locally or efficiently.
The law of comparative advantage, therefore, states that a country should concentrate on producing those goods in which it has the greatest relative cost advantage and imports from other countries those goods in which it has the greatest relative cost disadvantage.

2.3 THEORY OF ABSOLUTE ADVANTAGE

Absolute advantage means that a country is more efficient in the production of both goods under consideration than the other country being considered. A country's absolute advantage is measured in relation to other nations. If two countries are producing the same product, the country that produces the product cheaply has an absolute advantage over the other.

Even if a country has absolute advantage over the other in all products, there is still a possibility for the two nations to trade as illustrated in the example below.

For example, if two nations produce computers and cars as follows:

	Computers	Ratio (calculate opportunity cost)
Country X	10, 000, 000	10:1 in favour of country X
Country Y	1,000,000	
	Cars	
Country X	1,000,000	2: 1 In favour of country X again
Country Y	500,000	

Given the scenario above, it is still possible for the two countries to engage in trade. Absolute advantage cannot hamper international trade.
Country X has absolute advantage in both cases but the size of its advantage is greater in computer production than in car production. Country Y's disadvantage is smaller in car production than in the computer production. So it would be more beneficial to both countries if country X specializes in computer production and imports cars, while country Y specializes in cars production and imports computers.

3.0 FREE TRADE AND ITS EFFECTS

Free trade is a situation whereby the flow of goods, services and capital are not hindered by any artificial barriers. In theory, trade on an international level, should be free from any restrictions, and those who advocate for free trade maintain that this, would lead to numerous advantages, such as

- Enabling countries to specialise and increase production bearing in mind that the surplus can be exported.
- Countries export surpluses and import what they lack.
- Access to the world market, therefore enabling countries to benefit from economies of scale.
- Allowing countries to develop their industries as a result of free movement of capital.
- Promoting beneficial political links and closer cooperation between countries.
- Increasing efficiency due to competition from imports and limiting the creation of monopolies.
- The efficient use of resources also leads to lower costs of production which in turn leads to the reduction in the prices of goods and services.
- Provision of goods that were previously unavailable, a wider choice of goods to consumers.

3.1 DISADVANTAGES OF FREE TRADE

- It leads to unemployment especially in cases where imported goods are subsidised by the countries of their origin.
- It has negative effects on new industries.
- Dumping of imports on the local market leads to unfair competition.
- It may lead to the importation of undesirable products.
- The government will lose revenue because it can longer impose taxes on imports.

4.0 BARRIERS TO INTERNATIONAL TRADE

In spite of the numerous advantages of international trading, countries the world over engage in some form of **protectionism**. There are different forms of protectionism, and some of them are:

- **Quotas**. These are limits imposed on specified goods to be brought in the country. Import quotas restrict the quantity of certain products, which can be imported into the country. If the product is homogeneous then a simple quota is imposed. If they are heterogeneous, then the quota can take the form of a value of imports allowed in any given currency.

The effect of quotas is to reduce the volume of imports, raise the price of imports and encourage the demand for locally produced commodities.

Note that sometimes one country persuades another country to voluntarily reduce its exports of a product to a certain acceptable level, this is known as **voluntary export restraints (VERs).** VERs is also known as orderly market arrangements emphasizing their negotiated manner. VERs often apply to key industries, an example is VERs negotiated by the United States of America on Japanese exports of motor vehicles.

- **Tariffs or custom (import) duties.** These are taxes that are levied on imports. It can be a fixed amount per unit (specific) or a percentage of the price (ad valorem).

  The effect of tariffs is to raise prices of imports, and therefore reduce their demand, encourage the demand for locally produced commodities, as well as raise revenue for the government.

- **Trade embargoes.** This is a complete ban of imports from a particular country. Sometimes it is a total ban imposed on particular products like drugs, from any country! During the Iraq war of the early 1990s, the United Nations imposed a ban on Iraq's exports.

- **Hidden export subsidies and import restrictions (Direct controls).** This is a range of government subsidies and assistance for exports and deterrents against imports as follows:

- **Subsidies**. The government gives subsidies to local firms to allow them to compete favourably in terms of pricing of goods, with foreign firms.
- **Export credit guarantees** or insurance against bad debts for overseas sales.
- **Grants** or any form of financial help is provided to firms in the export sector
- **Zero rating** or reducing taxes on exported goods
- **State assistance** provided for firms in the export sector via the foreign office.

In addition, imports are discouraged through

- **Health and Safety regulations**. Countries sometimes put in place health and safety regulations that limit the importation of certain goods. For example, the Zambian government has put in place a regulation that stipulates that sugar sold in Zambian market must be fortified with vitamin A regardless of whether this sugar is locally produced or imported.
- **Administrative procedures (bureaucracy).** These are long, complex and costly procedures that importers have to go through at border posts.

- **Exchange controls.** These are aimed at restricting the amount of foreign exchange that is available to importers.

4.1 ARGUMENTS IN FAVOUR OF PROTECTIONISM

a) **To protect new and declining industries**.
New industries need to be protected from foreign competition before they become strong to be on their own, while declining industries might quickly collapse and lead to mass unemployment if not protected.

b) **To reduce unemployment**. Unfair competition from foreign products may lead to the closure of home industries. Therefore, the government protects its industries in order to prevent the closure of industries and unemployment.

c) **To reduce or eliminate balance of payments deficits.** Restricting imports will help to reduce or eliminate balance of payments deficits.

d) **To raise revenue.** The government raises revenue from import tariffs that are imposed on imported products.

e) **To protect strategic industries.** Industries such as ship building, defence and aerospace are of strategic importance to many countries. Therefore many countries protect these industries from foreign competition

f) **To protect against dumping of imported products on local market**. Dumping is a situation where goods are sold at lower prices in a foreign market than in the home market.

g) **Retaliation** against measures taken by another country that are unfair.

h) **To prevent unfair competition**. Governments may justify protectionism with reference to the trading policies of its competitor nation, such as selling imitations at artificially low prices.

4.2 ARGUMENTS AGAINST PROTECTIONISM

a) **The fear of retaliation**. If a country imposes restrictions against other countries' exports, the affected countries can retaliate by imposing restrictions on its exports.

b) **Reduction in industrial efficiency.** Protecting industries from international competition reduces their efficiency.

c) **Cost to consumers**. Protectionism is costly to consumers because they are forced to pay high prices for goods of poor quality.

d) **Restricted choice of products** Protectionism leads to the reduction in the range of products available to customers.

5.0 TERMS OF TRADE

The terms of trade are the ratio of an index of (visible) exports prices to an index (visible) import prices. They measure the relative change of the price of exports and the price of imports.

A base year is chosen, at which point the average price of exports is assigned an index value of 100, as is the average price of imports.

Suppose that in 2000, the base year, the average price of a basket of visible Zambian exports was K450, 000, while the average price of a basket of imports was K500, 000. Each of these prices would be assigned an index of 100, and the terms would be (100/100) = 1.

In fact, 100 multiply this ratio when the terms of trade are calculated. So the terms of trade for the year 2000 are 100.

Now suppose that in the following year (2001) the average price of exports rose by 10%, to K495, 000, while the average price of imports rose by 6%. The index for exports would rise to 100x 1.1 = 110, and the index for imports would rise to 106 (100x1.06). The terms of trade for 2001 would be (110/106) x 100 =104.

The rise in the terms of trade reflects the fact that export prices have risen more than import prices. An increase in the terms of trade is called an improvement in the terms of trade, though it may not always be desirable.

One reason for wanting an increase in the terms of trade is that a given quantity of exports will now pay for more imports. In the example above, the foreign currency earned by exporting one basket of exports in the year 2000 (K450, 000 worth) would buy 450/500 =0.9 or 90% of a basket of imports.

When the terms of trade improved in 2001, so that the average price of exports was K495, 000, while that of imports was K500, 000 x1.06 = K530, 000, one basket of exports would earn enough foreign currency to pay for 495/530 = 0.9339 or 93.34 % of a basket of imports. This means that, ceteris paribus, fewer exports are required to pay for imports.

An improvement in the terms of trade will be advantageous if it results in an increase in funds coming in and/ or a decrease in funds leaving the country. This happens if the **PED** for imports and exports is less than 1.

6.0 REGIONAL AND INTERNATIONAL ORGANISATIONS

6.1 THE SOUTHERN AFRICAN DEVELOPMENT COMMUNITY (SADC)

▪ **HISTORICAL BACKGROUND**

The Southern African Development Community (SADC) came into existence in 1980, as an alliance of independent Southern African States. The Southern African Development Community (SADC) was formerly known as the Southern African Coordination Conference (SADCC) and formed with the goal of helping countries in the Southern African region to lessen dependence on South Africa. The SADC headquarters are in Gaborone, Botswana.

▪ **MEMBER STATES**

The member states are Angola, Botswana, the Democratic Republic of Congo, Lesotho, Madagascar, Malawi, Mozambique, South Africa, Swaziland, Tanzania, Zambia and Zimbabwe.

▪ **OBJECTIVES OF SADC**

- The achievement of economic growth and development in member countries.
- Promotion of peace and security
- Promotion of common political beliefs and values
- Promotion of self-sustaining development
- Achievement of self- sustaining utilisation of natural resources and protection of the environment.
- The strengthening of long standing cultural links among the peoples of the SADC region.

▪ **ACHIEVEMENTS**

- Increase of trade among member sates.
- Member countries have strengthened their bargaining power.

▪ **CHALLENGES**

- Each member state uses a different currency and this hampers free trade.
- Over dependence on donor funding.
- Lack of participation in decision-making and other SADC activities by ordinary citizens.

▪ **FUTURE OUTLOOK**

SADC faces a bright future due to the following reasons:

- Donor confidence has been increasing.
- The SADC market is big with a population of 60 million people and rich in mineral resources.
- The region is likely to attract more investment with attainment of peace in Angola and the Democratic Republic of Congo (DRC).

6.2 COMMON MARKET FOR EASTERN AND SOUTHERN AFRICA (COMESA)

▪ HISTORICAL BACKGROUND OF COMESA

COMESA was established in 1993 in Uganda replacing the Preferential Trade Area (PTA) that was founded in 1981. Its headquarters are in Lusaka, Zambia.

▪ MEMBER STATES

These are Angola, Burundi, Comoros, Egypt, Eritrea, Ethiopia, Kenya, Lesotho, Madagascar, Malawi, Mauritius, Mozambique, Namibia, Rwanda, Sudan, Swaziland, Tanzania, Uganda, Democratic Republic of Congo (Zaire) Zambia and Zimbabwe.

▪ OBJECTIVES OF COMESA

- To achieve sustainable economic and social progress in all member countries.
- To promote economic cooperation among member states.
- To establish and maintain a Free Trade Area.
- To remove all tariff and non- tariff barriers.
- To create a Customs Union
- To promote free movement of capital and investment.

▪ ACHIEVEMENTS

- Creation of the COMESA Free Trade Area (FTA).
- Creation of an enabling environment for trade.
- A wider, harmonised and more competitive market.
- Greater industrial productivity and competitiveness.
- Increased agricultural production and food security
- More harmonised monetary, banking and financial resources.

▪ CHALLENGES

- Inability to participate effectively in the World Trade Organisation (WTO) negotiations.
- Resistance to elimination of trade barriers
- Lack of political stability in some member states like the Democratic Republic of Congo (DRC).
- Difficult to co-ordinate countries of vastly different economic, social and political backgrounds.
- Undeveloped infrastructure such as roads and telecommunication networks in some member states.

▪ FUTURE OUTLOOK

COMESA faces a bright future for the following reasons:

- COMESA’s population of over 300 million people constitutes a potentially large market and a huge reservoir of both skilled and unskilled labour.
- The COMESA region covering an area of about 12.89 million square kilometres is rich is minerals, lakes and rivers that can be exploited for irrigation, hydroelectric power and fisheries.
- Increased regional integration in trade and investment will lead to an expansion of the industrial and services sectors of member states.

6.3 THE EUROPEAN UNION (EU)

- **HISTORICAL BACKGROUND OF THE EU**

The European union (EU) formed in 1992 is an intergovernmental union of 25 countries of the European continent. Its headquarters are in Brussels, Belgium.

- **MEMBER STATES**

The EU is made up of 25 countries namely Italy, United Kingdom, France, Germany, Greece, Luxembourg, Netherlands, Ireland, Portugal, Spain, Belgium, Sweden, Finland, Denmark, Austria, Cyprus (Greek Part), Czech, Estonia, Hungary, Latvia, Lithuania, Malta, Poland, Slovakia and Slovenia.

- **OBJECTIVES OF THE EU**

- Abolition of remaining controls on capital flows.
- Removal of all non- tariff barriers to trade.
- Progress in harmonising tax rates.
- Removal of frontier controls and bias in public sector purchasing to favour domestic producers.

- **ACHIEVEMENTS**

- Creation of a single market consisting of customs union, a single currency managed by the European Central bank.
- Establishment of a common policies in agriculture, trade, fisheries and foreign and security.
- Abolition of passport control and custom checks at many of EU’s borders.
- Creation of single space of mobility for EU citizens to live, travel, work and invest.

- **CHALLENGES**

- Adoption, abandonment or adjustment of the new constitutional treaty.
- The EU’s enlargement to the south and east.
- Resolving of the EU’s fiscal and democratic accountability.
- Economic viability with the United States, China and India.

6.4 THE WORLD TRADE ORGANISATION (WTO)

▪ HISTORICAL BACKGROUND OF WTO

The World Trade Organisation (WTO) was formed in 1995. It replaced the General Agreement on Trade and Tariffs (GATT) that was established in 1948. Its secretariat is based in Geneva, Switzerland. The main purpose of WTO is to promote free trade by persuading countries to abolish import tariffs and other barriers.

▪ MEMBER STATES

In November 2005, membership of the WTO stood at 149 countries.

▪ OBJECTIVES OF THE WTO

- To promote free trade by persuading nations to abolish import duties and other barriers.
- To oversee the rules of international trade.
- To police free trade agreements.
- To settle trade disputes between member countries.
- To organise trade negotiations.

6.5 THE WORLD BANK (THE INTERNATIONAL BANK FOR RECONSTRUCTION AND DEVELOPMENT)

The World Bank came into existence in 1945 but commenced its operations in 1946. It is a non-profit organisation owned by member governments and has its headquarters in Washington, D.C in the United States of America.

OBJECTIVES OF THE WORLD BANK

- To fight poverty and improve the living standards of people in developing countries.
- To provide long-term loans and grants to developing countries.
- To provide technical assistance to help developing nations in their quest to reduce poverty.
- To provide assistance to developing countries on issues of economic development.

6.6 THE INTERNATIONAL MONETARY FUND (IMF)

The IMF was formed in 1944 by the Bretton Woods Agreement and started operating in 1947. The principal function of the IMF is to help countries with balance of payment problems. Its headquarters are in Washington, D.C in the United States of America.

OBJECTIVES OF THE IMF

- To promote international monetary cooperation and international payments.
- To encourage international trade among member states.
- To promote exchange rate stability in member states.
- To eliminate or remove of foreign exchange restriction
- To provide advisory services to member states
- To ensure that there is adequate supply of international liquidity.

7.0 CHAPTER SUMMARY

International trade is important because it allows countries to specialize according to the law of comparative advantage.

The law of comparative advantage states that international trade is most efficient and advantageous if each country sells the goods of which the country has the most advantage in production relative to other goods received in exchange.

There are a number of advantages of international trading such as giving consumers in each country more choice, encouraging efficiency in production, which is likely to result in lower prices.

In spite of the numerous advantages of free trade, countries engage in protectionist policies for various reasons. These include protecting employment, helping infant industries, preventing unfair competition, protecting the balance of payments, and raising revenue.

There are some arguments against protectionism. It is argued that they encourage inefficiency, lead to misallocation of resources, raise the cost of living, and retaliation may occur.

The methods or forms which protectionism takes include tariffs, quotas, hidden import restrictions and export subsidies.

The terms of trade refer to the rate at which exports can be exchanged for imports. An improvement in the terms of trade may not necessarily be beneficial as it reflects an increase in export prices arising from domestic inflation. However, an improvement in the terms of trade does reflect that fewer exports need to be sold to pay for each import because export prices are rising faster than import prices.

There are a number of international institutions, which facilitate international trading. Some of the international institutions are regional groupings that can take different forms, such as free trade areas, customs union and/or common markets.

CHAPTER 8

BALANCE OF PAYMENTS AND EXCHANGE RATES

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After studying this chapter, the students should be able to:

- ➢ Explain the monetary aspects of international trade transactions
- ➢ Understand the composition of the balance of payments account
- ➢ Explain how to correct a deficit balance of payments
- ➢ Appreciate of different types of foreign exchange systems
- ➢ Discuss the merits and demerits of each exchange rate system.
- ➢ Explain the difference between foreign exchange depreciation and appreciation
- ➢ Understand factors influencing foreign exchange rates

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1.0 INTRODUCTION

Balance of payments (BOP) is a measure of the payments that flow into and out from a particular country from other countries for a specific period usually a year. It is a statistical 'accounting' record of the international trading and capital transactions that have taken place during that year. This is determined by a country's exports and imports of goods, services, financial capital and financial transfers, and since buying goods and services from foreign countries is complicated by the fact that countries use different national currencies, this last chapter also deals with the foreign exchange market.

1.1 COMPOSITION OF BALANCE OF PAYMENTS

The Balance of payments consists of two parts namely

(a) **The current account.** This shows a record of net flow of money from transactions involving the purchase of goods and services, and transfer payments. The current account itself is divided into basically two parts namely

i) **Trade in goods** (known as **visible** or **trade** account)
The **exports** and **imports** of physical commodities such as copper and maize are recorded on this account. The account may show a surplus or a deficit. Exports are money coming into a country and if the exports are higher than the imports, then there is a surplus on thevisible trade account and vice versa.

ii) **Trade in services** (known as **invisible trade** account)
This is usually recorded as a net figure, implying the difference exports and imports of

services such as tourism, insurance, civil aviation, patents, foreign aid, grants, gifts etc.

It also includes the difference between **factor incomes** payable and receivable such as wages to foreign workers, interest on foreign debt, dividend payments on shares held by foreigners or income on any foreign investments. It is recorded as net transfer income from abroad or paid abroad depending on whether there is a higher net inflow or a higher net outflow of money respectively. This account can either be in surplus or deficit.

The current account, being a combination of the two accounts above, can either be positive or negative, that is either a surplus or a deficit. When exports or money received from trade in goods, services and factor incomes exceed imports or money paid for trade in goods, services and factor incomes, it means there a surplus or a favourable current account balance. When imports exceed exports, it means there is a deficit or an unfavourable current account balance. Note that this is what is referred to as the surplus or deficit balance of payments account!

(b) **The capital account.** This account records all international financial transactions in the country,the external assets and liabilities. In general it records medium and long-term capital inflows and outflows, including official reserves.

The **inflows** into the capital account:

- Foreign loans
- Investment by foreigners into Zambia
- Aid from donor countries
- A reduction in external reserves
- Trade in shares in Zambian investments by people based outside Zambia.
- Selling of assets that are based in other countries.

The **outflows** from the capital account:

- Zambians investing in other countries.
- Zambia giving aid or loans to other countries.
- Trade in shares by Zambians in investments abroad
- An increase in external reserves.
- Selling of assets based in Zambia to people based outside Zambia.

In summary, if foreign ownership of domestic assets has increased more quickly than domestic ownership of foreign assets in a given year, then the domestic country has a **capital account surplus**. On the other hand, if domestic ownership of foreign assets has increased more quickly than foreign ownership of domestic assets, then the domestic country has a **capital account deficit**.

The capital account ‘finances’ or ‘covers’ the current account, since a surplus or deficit on the current account must be ‘balanced’ by a deficit or surplus on the capital account respectively. If for example there is a negative or a deficit current account balance, then it must be ‘financed’ by inflows into the capital account and vice versa.

The above means the sum of the balance of payments account must always be **zero**. In practice, an additional account is included to achieve the zero balance. This is known as **net errors and omissions** or **a balancing item.**

1.2 Official Reserve Account

The official reserve account records the government's current stock of **reserves**. Reserves include official gold reserves, foreign exchange reserves, and strategic defense reserves (SDRs), such as the Strategic Petroleum Reserve.The Balance of Payments is the sum of the Current Account and the Capital Account. Therefore, **Balance of Payments = Current Account + Capital Account = *Change* in Official Reserve Account**! The balancing item is a change in the official reserves. A negative sign is an increase in official reserves while a positive sign is a decrease in official reserves.

1.3 Zambia’s balance of payments (extracts from the B.O.Z annual report)

The improvement in the terms of trade, on account of the increase in the international price of copper, as well as the attainment of the enhanced HIPC Completion Point contributed to the improvement in the performance of the external sector during the year. Consequently, the deficit in the overall balance of payments declined. Preliminary information indicates that the deficit in the overall balance of payments narrowed by 21.3% to US $274 million in 2005 from US $348 million in 2004. This improvement in the overall balance was largely due to the favourable performance in the capital and financial accounts despite the deterioration in the current account. Project loans and grants, foreign direct investment as well as portfolio inflows registered significant increases, and enhanced the capacity to build-up Gross Official International Reserves (GIR) of the Bank of Zambia. This development reinforced the positive effects of the improvement in the terms of trade.

Current Account

The combination of continued strong economic activity and the appreciation of the Kwacha led to an increase in the current account deficit through increased imports. There was a decrease in the merchandise trade balance as well as the deterioration in net services and income accounts. The merchandise trade balance declined by 28.0% to US $59 million in 2005 from US $82 million in 2004 while the net services and income accounts deteriorated by 17.2% and 42.7% to US $252.0 million and US $605.0 million, respectively. The decline in the trade balance resulted from a higher increase in the value of merchandise imports than that of merchandise exports.

The value of imports increased by 19.7% to US $2,068 million from US $1,727 million recorded in 2004. The increase in the import bill was in part explained by the continued high investment activity in the mining sector and the rise in the price of oil on the world market. The strengthening of the Kwacha against major trading currencies also reinforced increased domestic demand for imports.

TABLE 17: BALANCE OF PAYMENTS IN US $ MILLION, 2003 – 2005

	2003	2004	2005
Current account balance	**-700**	**-583**	**-826**
Trade balance	-311	82	59
Exports, f.o.b.	1,052	1,779	2,095
Metal sector	669	1322	1,557
Non-traditional	383	457	538
Imports, f.o.b.	-1,393	-1,727	-2,068
Metal sector	-169	-286	-306
Non-metal	-1,224	-1,441	1,759
Goods procured in ports by carriers	29	31	32
Services (net)	-238	-215	-252
Receipts	165	232	246
Payments	-403	-447	-499
Income (net)	-148	-424	-605
Of which: interest payments	-131	-121	-110
Current Transfers (net)	-3	-25	-28
Of which Official Transfers	20	0	0
Capital Account	**380**	**235**	**552**
Project grants (capital)	240	246	306
Financial Account	**140**	**-11**	**246**
Official loan disbursement (net)	-141	-221	-105
Disbursement	101	110	120
Amortization (-)	-242	-331	-225
Change in net foreign assets of Commercial banks	48	-90	17
Private capital (net)	233	299	334
Foreign direct investment	172	239	259
Errors and omissions, short term capital	-2	63	0
Overall balance	**-319**	**-348**	**-274**
Financing	321	285	274
Change in net international reserves of BoZ (-increase)	-161	-44	-341
Gross official reserves of BoZ	89	-28	-81

BoZ liabilities	-6	-6	-6
IMF (net)	-244	-10	-253
Debt Relief	389	264	480
Debt relief (non-HIPC)	154	245	152
Debt relief (HIPC, including IMF)	235	19	328
Of which IMF	169	2	229
Other Debt Related Items	-10	0	0
Net change in arrears (+ increase)	48	0	0
BOP support grants	45	44	105
BOP support loans	10	21	29
Multilateral	10	21	29
Bilateral	0	0	0
Financing gap (+)	0	0	0
Memorandum items:			
Nominal GDP (millions of US $)	4,326	5,422	6,968
Current account balance (% of GDP)	-16.2	-10.7	-11.9
Terms of trade (percentage change)	4.2	21.9	2.1
Copper volume (MT.'000)	353	393	417
Copper price (US$/lb)	0.78	1.20	1.52
Gross official Reserves	194	222	303
(In months of imports)	1.3	1.2	1.4
Debt service cash payments (US $m)	192	371	162
(In % of exports)	15.4	18.2	6.8
Of which; official debt service	108	114	129

Source: Bank of Zambia and Fund Staff Estimates
Notes: The figures reported in the Balance of Payments table are as estimated in October 2005

Total export earnings in 2005 increased by 17.8%, to US $2,095 million from US $1,779 million in 2004, with earnings of metal and non-traditional exports (NTEs) both rising by 17.8% to US $1,557 million and US $538 million, respectively. The increase in copper prices was mainly due to sustained international demand particularly from China and India while export volumes edged upwards as a result of continued recapitalisation of the existing mines and commencement of full production at Kansanshi mine.

In contrast, cobalt exports declined. The 17.7% increase in Non-Traditional Export was explained by growth in the exports of copper wire, sugar, burley tobacco, cotton lint and electrical cables.

The deficit in the services account deteriorated to US $252 million in the past year, reflecting large amounts of net payments made on trade-related services. With respect to the income account, the deficit widened by 42.7% to US $605 million. This increase was in spite of a 9.0%

decline in official interest payments on external debts to US $110 million from US $121 million over this period.

Capital and Financial Account

The surplus on capital and financial account rose to US $552 million in 2005 from US $235 million in 2004. This was largely due to increased donor inflows, foreign direct and portfolio investments as well as a reduction in debt amortization. Increased external capital inflows partly resulted from the rise in investor and donor confidence following the attainment of the enhanced HIPC Initiative Completion Point in April 2005.

Financing

The deficit in the overall balance of payments was financed mainly by debt relief of US $480 million, inflows of BoP support grants amounting to US $105 million and BoP support loans of US $29.0 million.

The Balance of Payments Support came from cooperating partners. Specifically, Zambia received from the European Union (EU), from the British Government under the Poverty Reduction Budgetary Support (PRBS) to finance priority poverty reduction programmes, from Finland, from Sweden, from the World Bank and from Norway.

2.0 THE PROBLEM OF BALANCE

When a country has an adverse or a deficit (negative) balance of payments, this is regarded with serious concern. When a country has a favourable or credit (positive) balance, then there is satisfaction.

Yet for all countries of the world, total payments must be equal to total receipts, since every payment is at the same time a receipt. Since all countries cannot achieve favourable balances in the same year, the aim should be an equilibrium balance of payments over a period of time.
Each individual country's balance of payments must balance each year. When all items have been taken into account, a balance is achieved by showing how the deficit or amount of credit (favourable) balance has been 'covered' or 'financed'.

If a country has a deficit in its balance, it must be 'covered' by **inflows** into the capital account. If it has a credit or positive balance, then it is 'covered' by **outflows** from the capital account.

2.1 CORRECTING A BALANCE OF PAYMENTS DEFICIT

- **Devaluation/depreciation**

Devaluation of a currency is a reduction in the exchange rate of the currency relative to other currencies. The objective of devaluing a country's currency to make exports cheaper and imports expensive, by reducing the price of exports to foreign buyers (i.e. in foreign currency terms) and increasing the price of imports in terms of the domestic currency.

If for example the Zambian Kwacha to the US dollar is devalued form $1 = K3200 to $1 = to K4000, then foreign consumers and firms will be encouraged to switch to Zambian goods because with the same $1, they are able to purchase more Zambian goods. They are able to purchase K4000 worth of goods instead of K3200 worth of goods. In addition, local consumers and firms will be discouraged from imports. They will need to have K4000 to purchase a $1 worth of goods, before devaluation, they needed to have K3200 to purchase a $1 worth of goods.

Therefore, depreciation in the kwacha exchange rate should help to boost the overseas demand for Zambian exports because Zambian firms will be able to supply more cheaply in international markets.

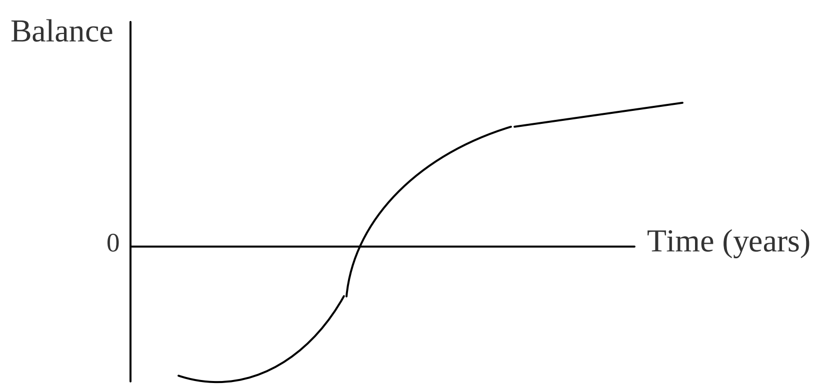
The extent to which export sales rise following a fall in the exchange rate depends on the price elasticity of demand for Zambian products from foreign consumers.

A lower exchange rate should also cause imports into Zambia to become relatively more expensive, thus leading to a slowdown in import volumes and "expenditure-switching" towards local goods. The significance of elasticity of demand is again important.

In the short run, even if elasticities are favourable, a depreciation/devaluation does not immediately benefit a balance of payments in practice. It takes time for movements in the exchange rate to affect trade flows. In the short run, demand for imports is likely to be fairly inelastic, while exporters would be unlikely to increase the output to meet the increase in demand due to the depreciation of the currency. Therefore, there is likely to be an initial worsening of the current account because volumes are fixed.

In the long run, demand and supply become more elastic, production and volume of exports rise, imports can be substituted and the volume of imports falls. This improves the current account balance.

The effect of devaluation/depreciation of a currency on the current account is called the j-curve effect. The exact shape of the curve depends on the assumptions made, and it is usually assumed that the improvements in the current account eventually levels off.



- **Deflation/Fiscal policy**

This is contraction of the domestic economy. Deflationary measures are aimed at reducing aggregate demand and this can be achieved by either increasing interest rate to discourage borrowing or increasing tax rates in order to reduce consumption expenditure. The government can also reduce its own expenditure.
Some of the overall trade deficit is due to the strength of domestic demand for goods and services. If and when the economy enters a slowdown phase, the growth of imports will fall, and this should provide an element of correction for the trade deficit

The effect of the deflationary measures is to reduce the demand for goods and services, including the demand for imports. If imports are high, demand for them is reduced by reducing the demand in the economy in general, as long as the demand for imports is income inelastic. If the fall in demand is accompanied by a reduction in inflation in the home market, the competitiveness of exports improves (as long as demand for exports in price elastic). In addition, firms are encouraged to switch to export markets because of the fall in domestic demand.

Some of the overall trade deficit is due to the strength of domestic demand for goods and services. If and when the economy enters a slowdown phase, the growth of imports will fall, and this should provide an element of correction for the trade deficit. The major problem associated with deflation is that a sharp fall in consumer spending might lead to a steep economic slowdown (slower growth of GDP) or a full-scale recession.

- **Direct controls (discouraging imports whilst encouraging exports).**

These are the direct controls mentioned earlier under protectionism. A government can impose trade restrictions like quotas, import duty, exchange controls, health and safety regulations etc. Increase exporters' competitiveness on the international market by subsidising exporters. A government may also adopt policies to promote exports e.g. zero-rating VAT on exports, export credit guarantees etc. Eventually the policies result in more exports.

The problem with this policy instrument is that there is a danger of other countries retaliation, as well as if the demand for imports is inelastic.

- **Raising interest rates.**

High interest rates are likely to make Zambia attractive to foreign investors and encourage inward investment, an **inflow** of foreign currency. This short- term capital movement of currencies by international financiers/speculators is known as 'hot money'. Higher interest rates act to slowdown the growth of consumer demand and therefore lead to cutbacks in the demand for imports. It is a short-term measure, which eventually leads to an appreciation of the exchange rate.
Note that the key to **long-term improvements** in trade performance is to focus on supply side policies. Controlling or reducing a balance of payments deficit in the long term is to achieve relatively low inflation with sufficient productive capacity to meet the domestic demand from consumers.

This requires a period of low inflation, low interest rates and a competitive exchange rate matched with sufficient non-price competitiveness in overseas markets. Price is not always the only deciding factor in winning the demand from buyers, investment in research and development and effective marketing strategies can have long term effects in maintaining market share.

An outward shift of long run aggregate supply would provide the economy with an increased capacity - permitting a reallocation of resources towards exporting. Therefore, a sustained improvement in the balance of payments requires businesses exploiting opportunities in export markets overseas, increased investment in services (including business finance, tourism) as many services are exportable and have the potential to earn huge sums in foreign currency. Improve efficiency and productivity in the export sectors.

3.0 EXCHANGE RATES

Exchange rates are the price of a currency expressed in terms of another currency. An exchange rate is the price for obtaining one unit of a foreign currency. The exchange rate refers to the value of one currency (e.g. the Kwacha) in terms of currency (e.g. the United States Dollar).

3.1 DETERMINATION OF EXCHANGE RATES

The basic forces behind the determination of exchange rates are those of supply and demand. Taking exchange rate for the kwacha against other currencies as an example, the exchange rate set in the market will be affected by supply of and demand for Kwacha.

3.2 DEMAND

People and firms want the Kwacha for various reasons:

a) To pay for Zambian exports.
b) Investors based abroad wanting to invest in Zambia.
c) Speculation. Speculators will buy the Kwacha at the current exchange rate, if they think it going to appreciate in the near future. They want to sell the Kwacha at a higher exchange rate in future.
d) The central bank may want to buy Kwacha to push up its value on the foreign exchange market.

3.3 SUPPLY

Supplies of Kwacha arise when people buy foreign currency in exchange for Kwacha. The factors affecting supply are as follow:

a) Zambian residents wishing to buy imports will require foreign currency, so they need the Kwacha to acquire foreign currency.
b) Zambian residents investing abroad will sell the Kwacha and buy foreign currency.
c) If speculators think that the Kwacha is about to depreciate they sell it.

d) The central bank may sell the Kwacha to manipulate its value.

3.4 FIXED EXCHANGE RATES

This is where the government keeps the exchange rate at a fixed level, but if it cannot control inflation, the real value of the currency will not remain fixed.

3.4.1 ADVANTAGES OF FIXED EXCHANGE RATES

a) They make international trade more stable because of the certainty to traders.
b) They make it possible for importers and exporters to predict profits.
c) They make investors more confident about investing in other countries.
d) Importers and exports can agree prices for future delivery without having to worry about potential losses through exchange rate movements.

3.4.2 DISADVANTAGES

a) They require substantial official reserves. The Central Bank requires a large pool of foreign currency to enable a prolonged period of intervention to support the exchange rate.

b) They complicate Economic co-operation among countries with different Economic objectives and policies

c) Fixed exchange rates can lead to capital flight or outflows of capital if interest rates are attractive in other countries.

3.5 FLOATING EXCHANGE RATES

Floating exchange rates are exchange rates that are determined by the market forces of supply and demand. Under the Floating exchange rate system, the government does not intervene in the foreign exchange market. A system under which exchange rates are not fixed by government policy but are allowed to float up or down in accordance with supply and demand.

3.5.1 ADVANTAGES

a) The nation's exchange rate will adjust automatically in the foreign exchange market to correct any balance of payments deficits or surplus.

b) The central bank does not need to large reserves maintain a certain exchange rate.

c) Monetary policy will be more effective.

d) There is no need to work out the new exchange rate because market forces of supply and demand will determine it.

3.5.2 DISADVANTAGES

a) Floating exchange rates lead to uncertainty in international trade and this may hinder trade with other countries.

b) Floating exchange rates encourages speculation, which in turn leads to increased volatility of the exchange rates.

c) Fiscal policy will be less effective.

3.6 Note the following in relation to exchange rates: In markets where exchange rates **float**, an **increase** in the external value of a currency is referred to as an **appreciation** and a **decrease** in the external value is referred to as a **depreciation** of the currency.

In markets where exchange rates are **fixed**, when authorities **raise** the external value of the currency to **a higher** fixed parity, this is referred to as a **revaluation** and a change to a **lower** parity is referred to as a **devaluation** of the currency.

3.7 MANAGED FLEXIBILITY OR DIRTY FLOATING EXCHANGE RATES

The system that exists in practice is a compromise between fixed and floating rates.

The market forces now play a more important role in the determination of exchange rates, but the authorities often intervene to neutralise short run pressure on exchange rates, to ensure that it remains fixed within a certain zone of flexibility. This system is known as a managed float.

A central bank, on behalf of the government, buys and sells currency to stabilize the exchange rate. When one reads in the newspapers that the Bank of Zambia has offloaded foreign currency (from its reserves) to buy the Kwacha on the foreign exchange market, the bank wants to artificially stimulate demand, and make the Kwacha appreciate in value, and vice versa.

However, because authorities do not always make it clear that they are using the reserves to support a currency's external value, and maintain an exchange rate target, which is usually unofficial, the term **dirty float** is used.

4.0 CHAPTER SUMMARY

The balance of payments is an account showing the financial transactions of one nation with the rest of the world, it records flows of funds between residents of a country and overseas residents, normally for a period of one year.

The balance of payments consists of two parts, the current account and the transactions in external assets and liabilities, known as the capital account.

The current account is made up of the visible and the invisibles account, while the capital account shows the inflows and the outflows of foreign currency. The overall balance shows how the difference between current and capital accounts is financed.

In theory, the balance of payments always balances because of the double entry system used in recording transactions. However, in practice, there is need to include a balancing item, which is created by errors and omissions in measuring the figures.

A balance of payments deficit indicates that what was paid out is greater than what was received. The deficit can be adjusted by devaluation and deflationary measures as well as direct controls.

A balance of payments surplus can be adjusted by revaluation, and other measures to encourage imports. Exchange rates are a ‘price’ of one unit of a currency expressed in terms of another currency.

There are two basic exchange rate systems, the flexible or the floating and the fixed exchange rate systems, in practice, most exchange rates fall in between these two extremes. A ‘dirty’ or managed exchange rate is when the central bank intervenes in order to stabilize the exchange rate.

The market forces of supply and demand for a currency determine the floating exchange rate, and the central authorities determine the fixed exchange rate system. Both systems have advantages and disadvantages. When there is a high demand for a currency due to an increase in exports or other factors, the currency appreciates in value and vice versa.